
Neural Machine Translation Decoding

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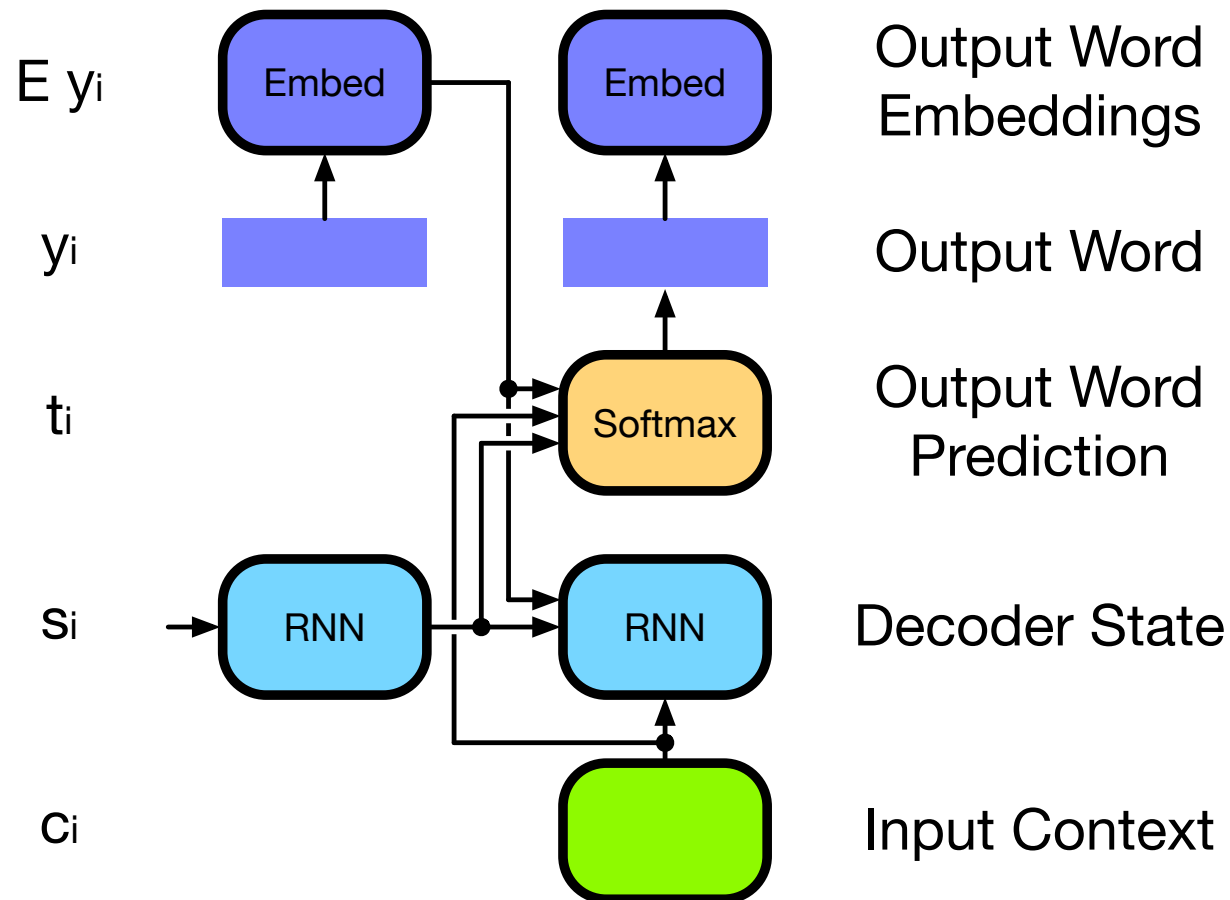


- Given a trained model
 - ... we now want to translate test sentences
- We only need execute the "forward" step in the computation graph

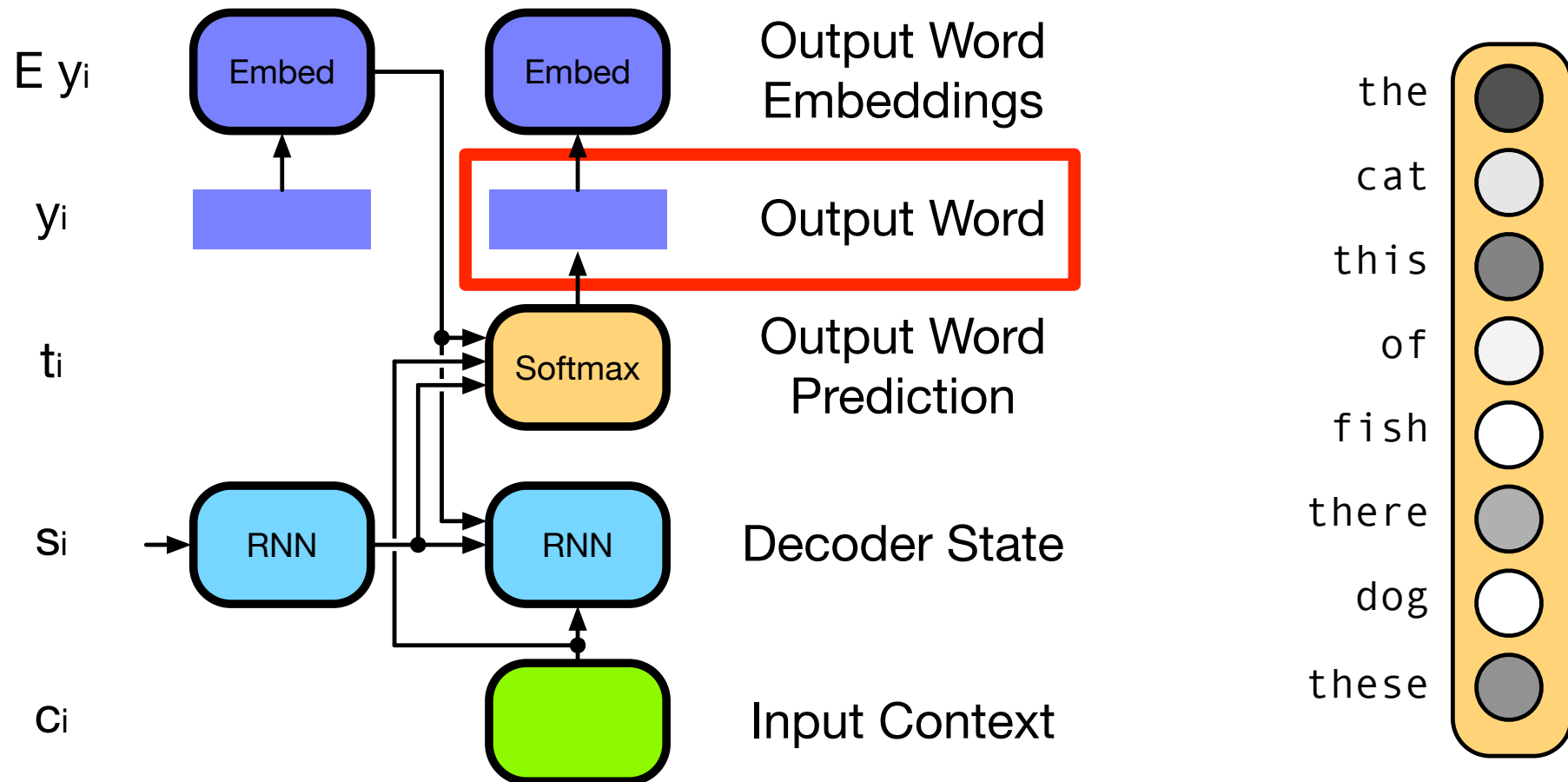
Word Prediction



2



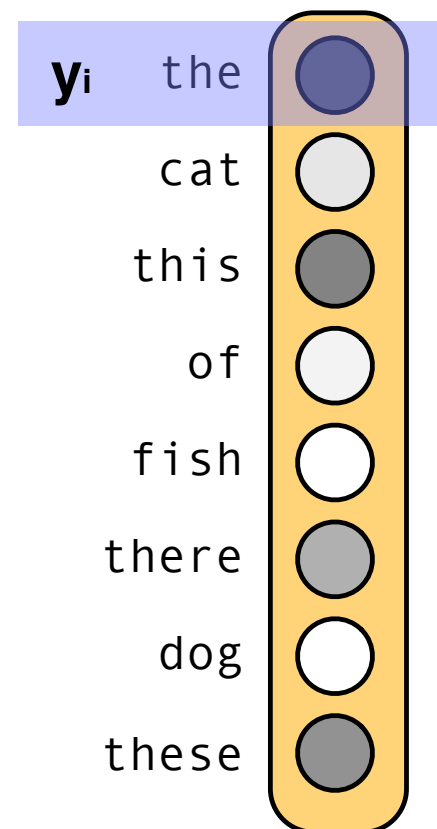
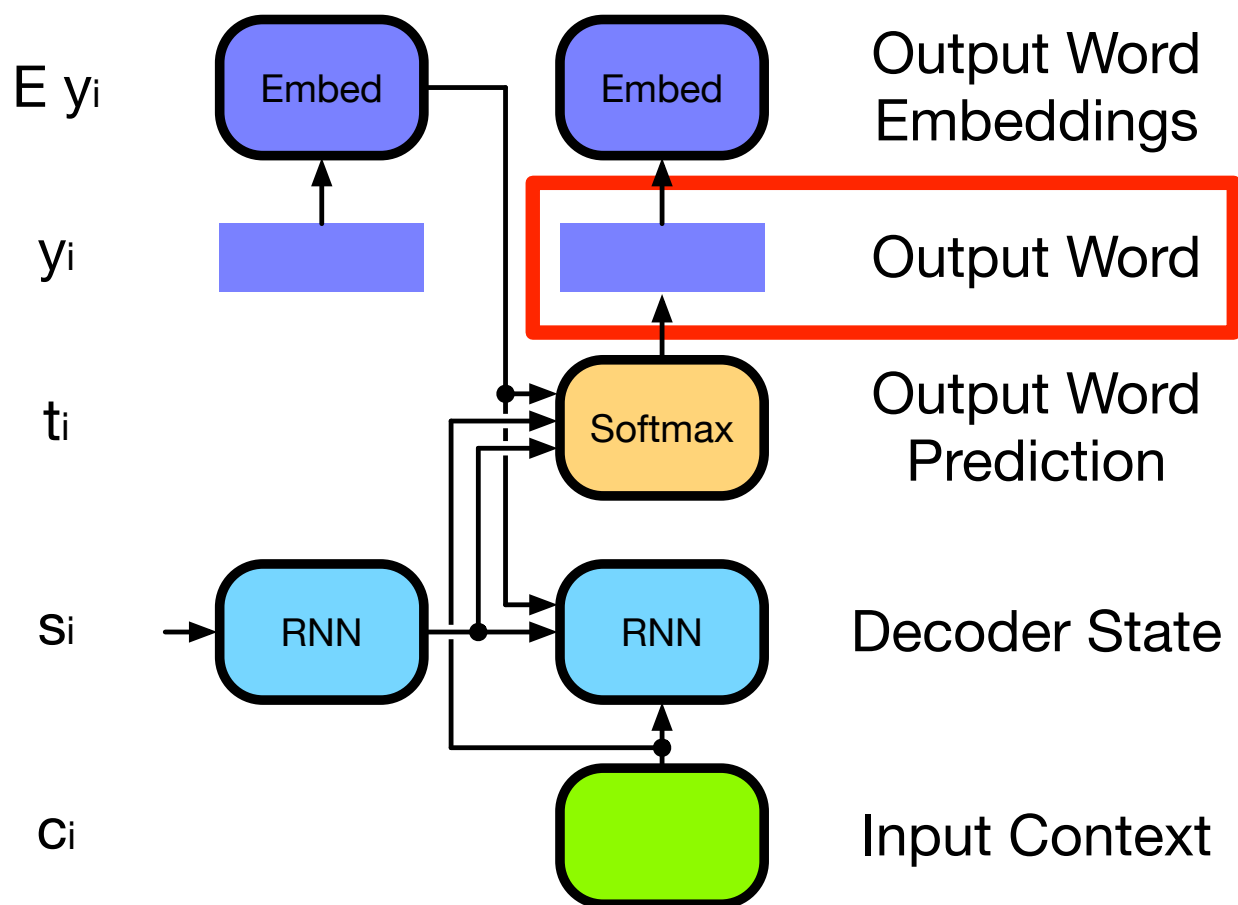
Selected Word



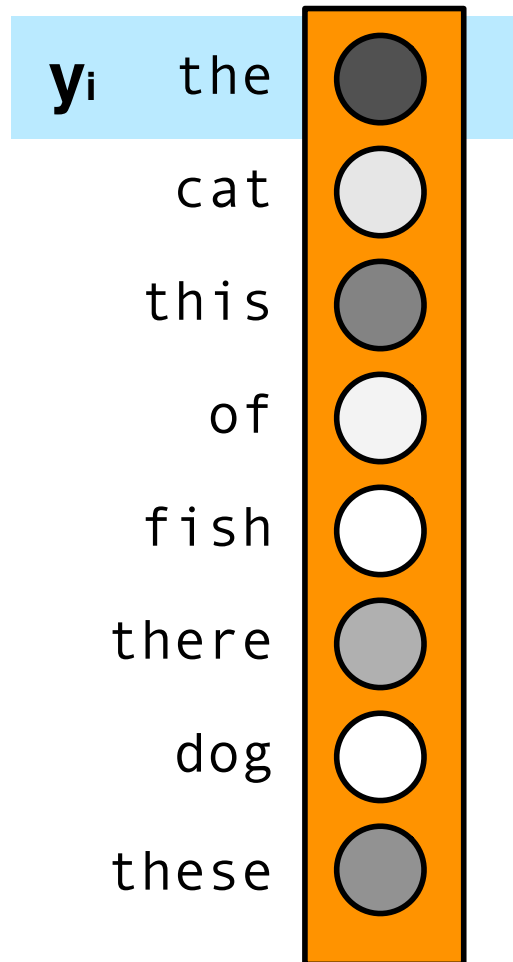
Embedding



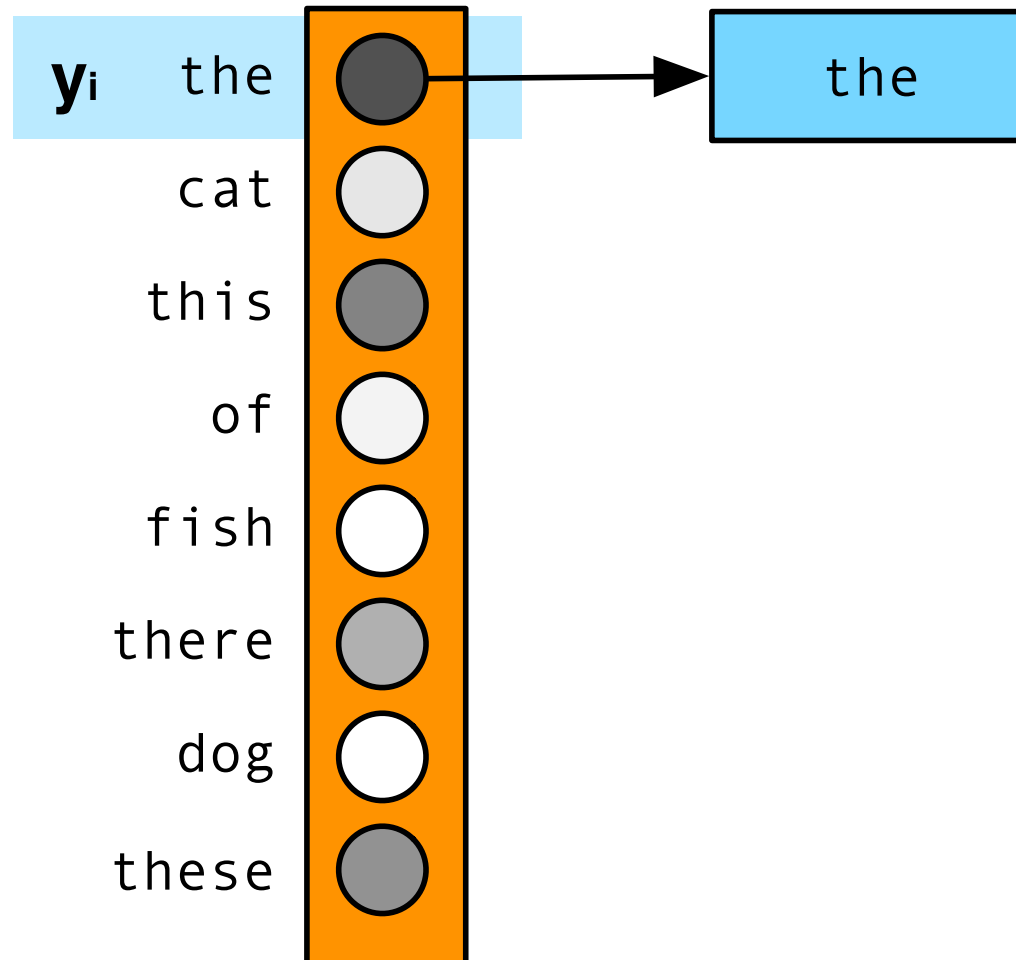
4



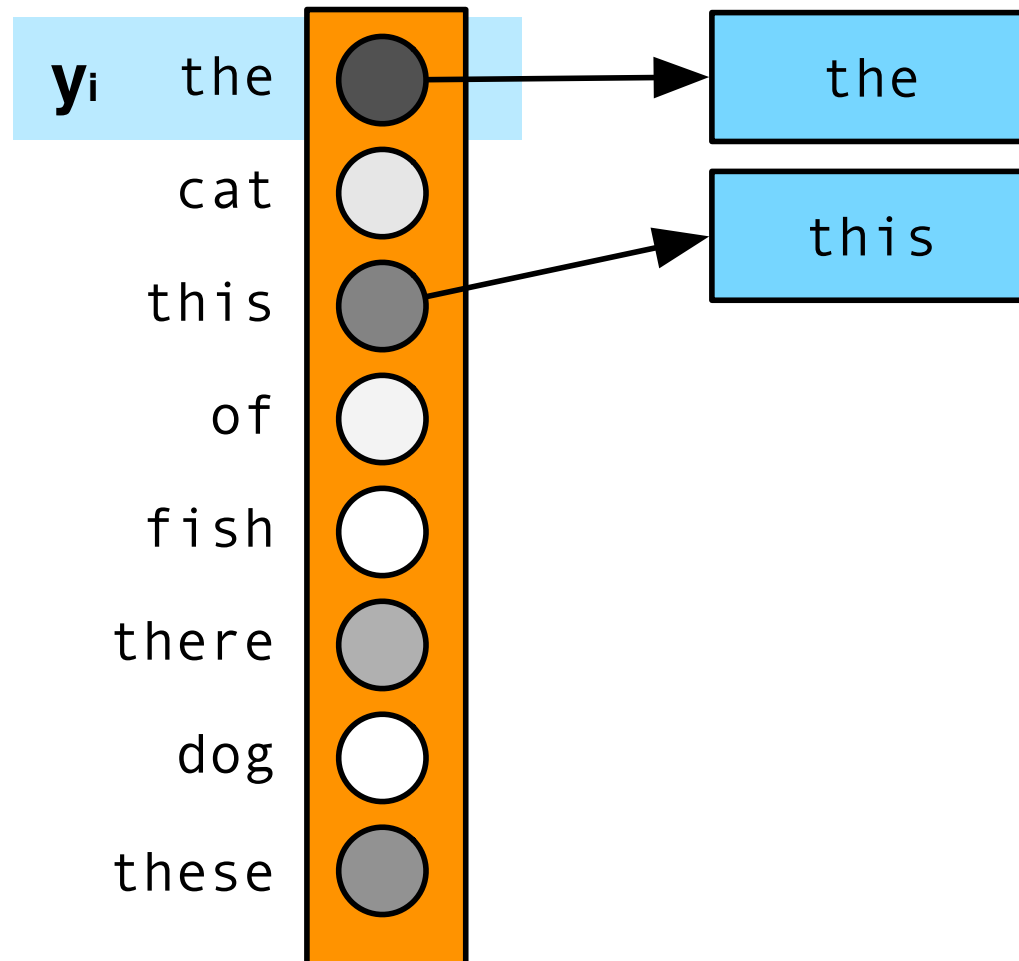
Distribution of Word Predictions



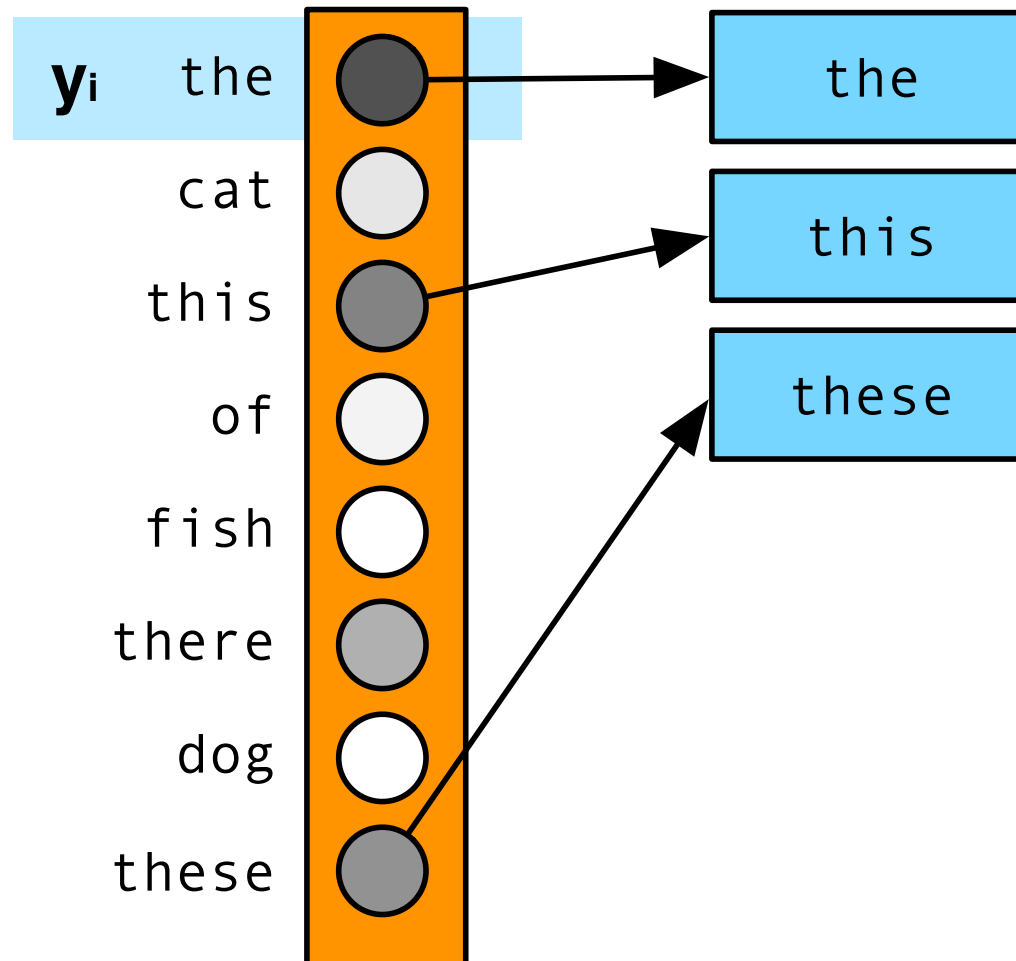
Select Best Word



Select Second Best Word



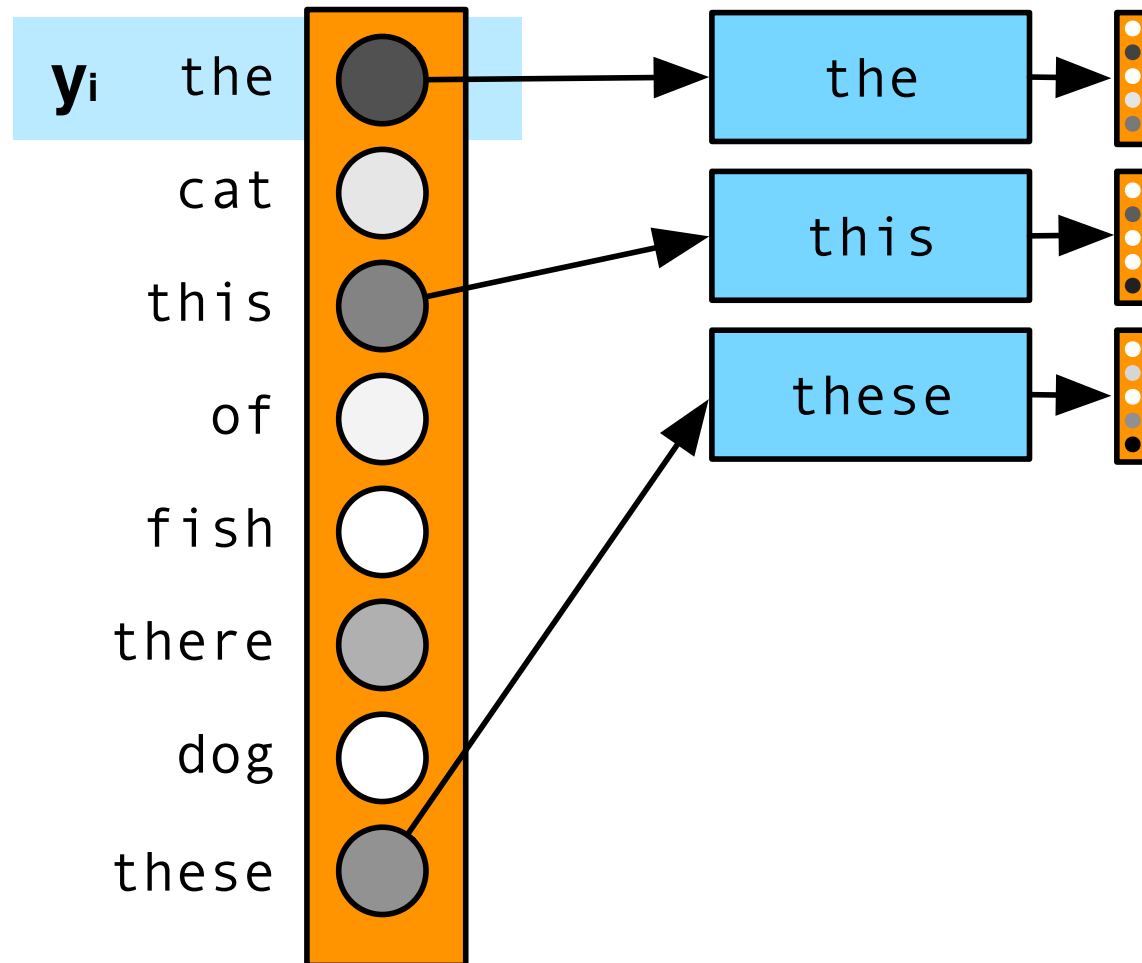
Select Third Best Word



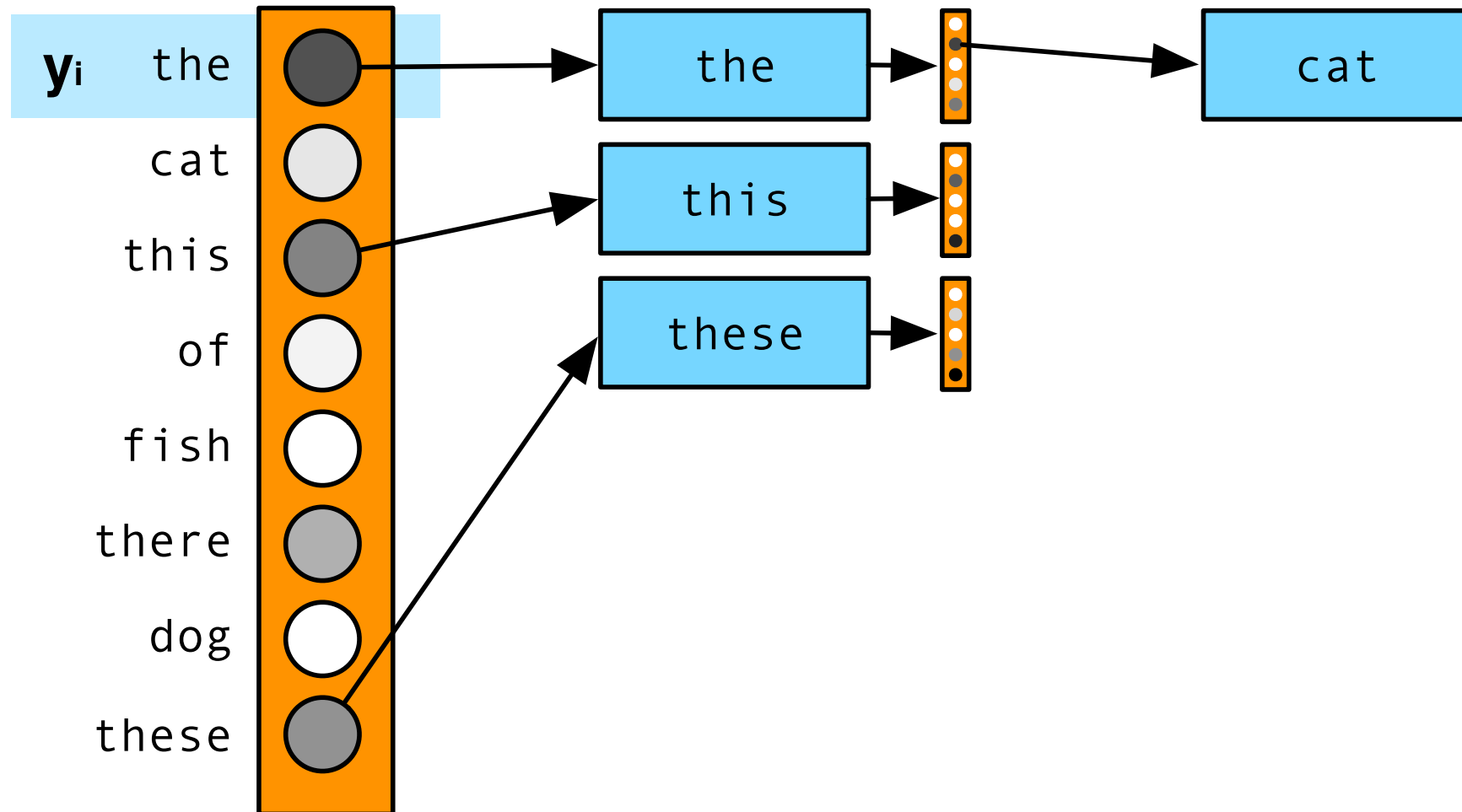
Use Selected Word for Next Predictions



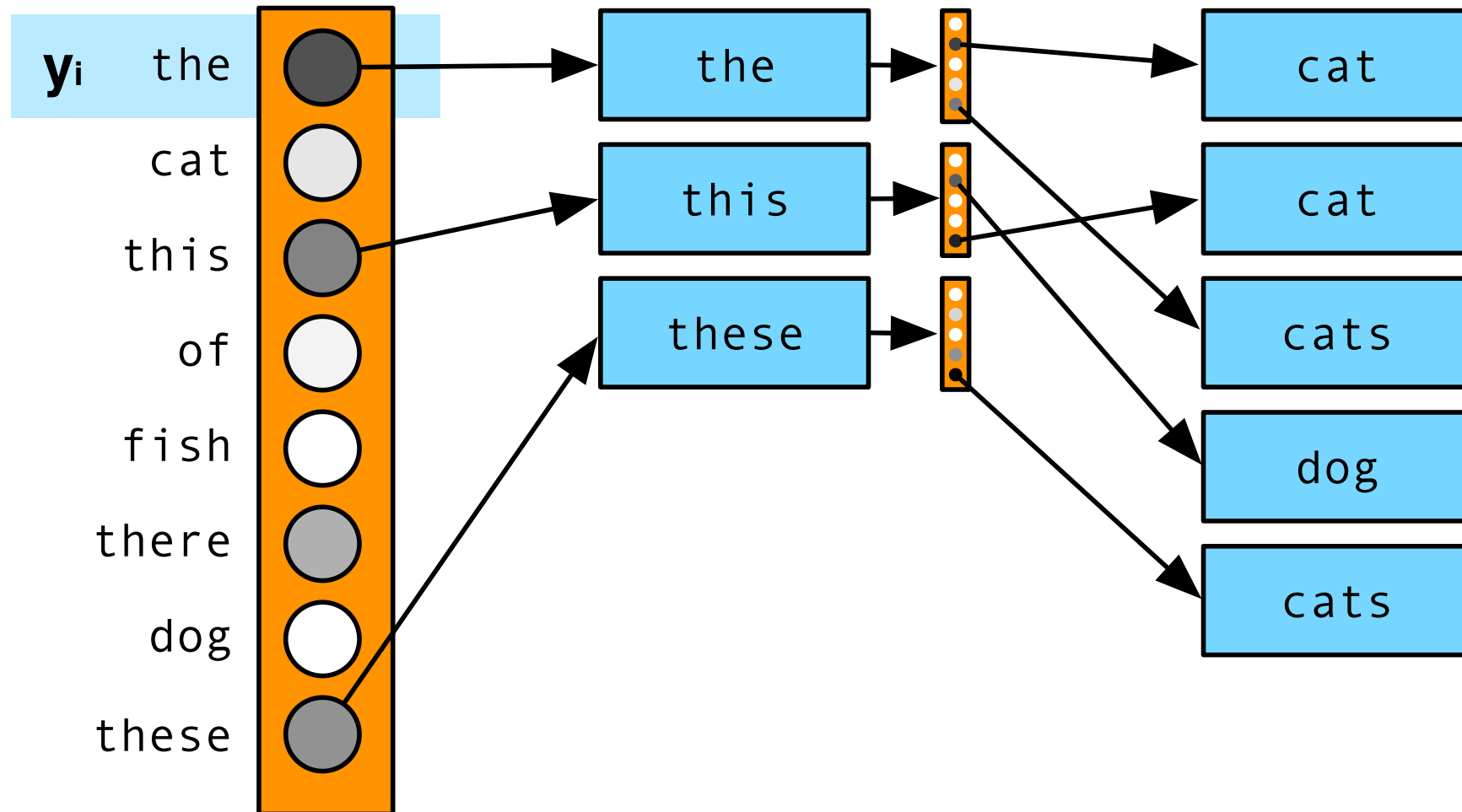
9



Select Best Continuation

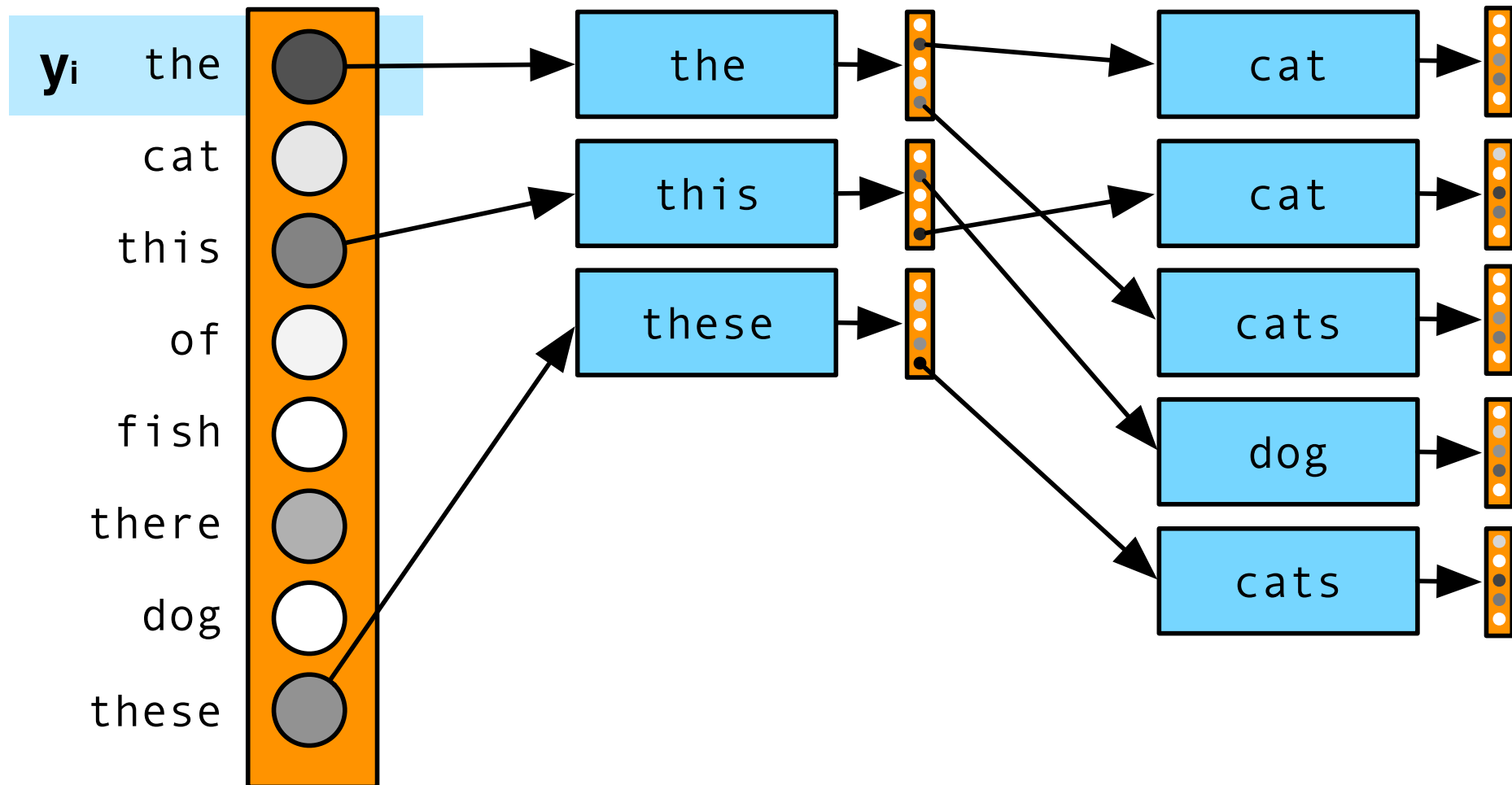


Select Next Best Continuations

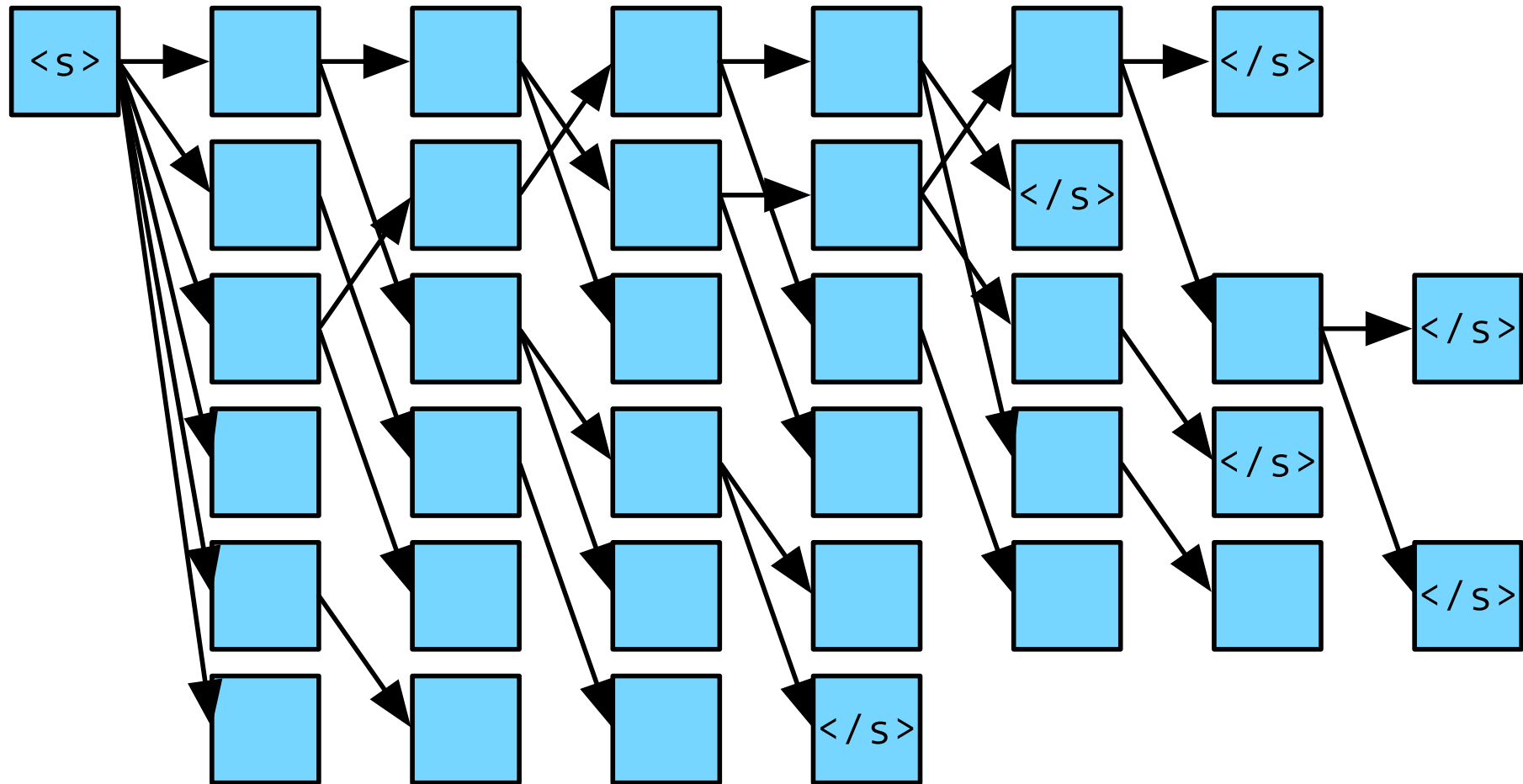


Continue...

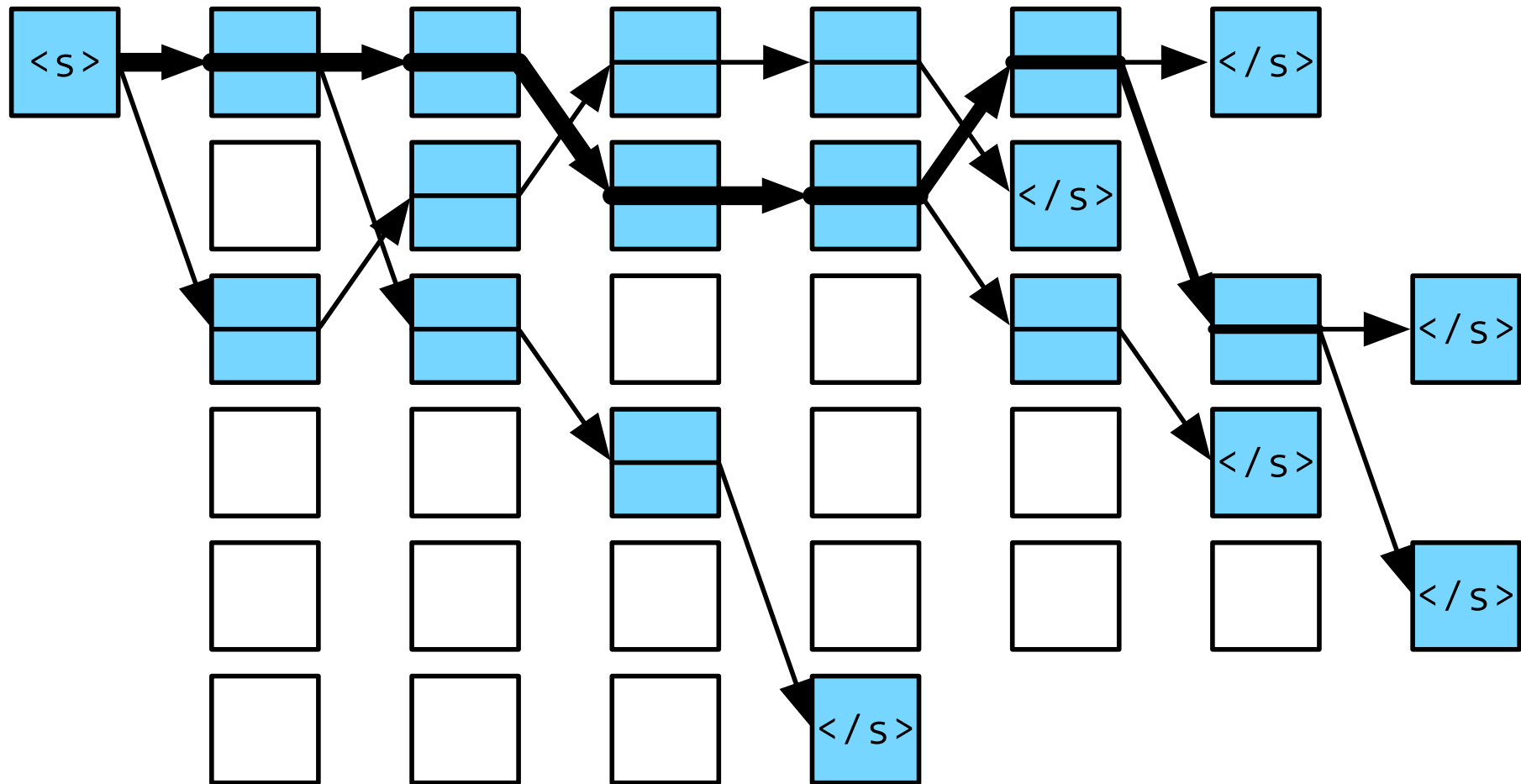
12



Beam Search



Best Paths



Beam Search Details

- Normalize score by length
- No recombination (paths cannot be merged)

Output Word Predictions

Input Sentence: *ich glaube aber auch , er ist clever genug um seine Aussagen vage genug zu halten , so dass sie auf verschiedene Art und Weise interpretiert werden können .*

Best		Alternatives
but	(42.1%)	<i>however (25.3%), I (20.4%), yet (1.9%), and (0.8%), nor (0.8%), ...</i>
I	(80.4%)	<i>also (6.0%), , (4.7%), it (1.2%), in (0.7%), nor (0.5%), he (0.4%), ...</i>
also	(85.2%)	<i>think (4.2%), do (3.1%), believe (2.9%), , (0.8%), too (0.5%), ...</i>
believe	(68.4%)	<i>think (28.6%), feel (1.6%), do (0.8%), ...</i>
he	(90.4%)	<i>that (6.7%), it (2.2%), him (0.2%), ...</i>
is	(74.7%)	<i>'s (24.4%), has (0.3%), was (0.1%), ...</i>
clever	(99.1%)	<i>smart (0.6%), ...</i>
enough	(99.9%)	
to	(95.5%)	<i>about (1.2%), for (1.1%), in (1.0%), of (0.3%), around (0.1%), ...</i>
keep	(69.8%)	<i>maintain (4.5%), hold (4.4%), be (4.2%), have (1.1%), make (1.0%), ...</i>
his	(86.2%)	<i>its (2.1%), statements (1.5%), what (1.0%), out (0.6%), the (0.6%), ...</i>
statements	(91.9%)	<i>testimony (1.5%), messages (0.7%), comments (0.6%), ...</i>
vague	(96.2%)	<i>v@@ (1.2%), in (0.6%), ambiguous (0.3%), ...</i>
enough	(98.9%)	<i>and (0.2%), ...</i>
so	(51.1%)	<i>, (44.3%), to (1.2%), in (0.6%), and (0.5%), just (0.2%), that (0.2%), ...</i>
they	(55.2%)	<i>that (35.3%), it (2.5%), can (1.6%), you (0.8%), we (0.4%), to (0.3%), ...</i>
can	(93.2%)	<i>may (2.7%), could (1.6%), are (0.8%), will (0.6%), might (0.5%), ...</i>
be	(98.4%)	<i>have (0.3%), interpret (0.2%), get (0.2%), ...</i>
interpreted	(99.1%)	<i>interpre@@ (0.1%), constru@@ (0.1%), ...</i>
in	(96.5%)	<i>on (0.9%), differently (0.5%), as (0.3%), to (0.2%), for (0.2%), by (0.1%), ...</i>
different	(41.5%)	<i>a (25.2%), various (22.7%), several (3.6%), ways (2.4%), some (1.7%), ...</i>
ways	(99.3%)	<i>way (0.2%), manner (0.2%), ...</i>
.	(99.2%)	<i></s> (0.2%), , (0.1%), ...</i>
</s>	(100.0%)	

sampling

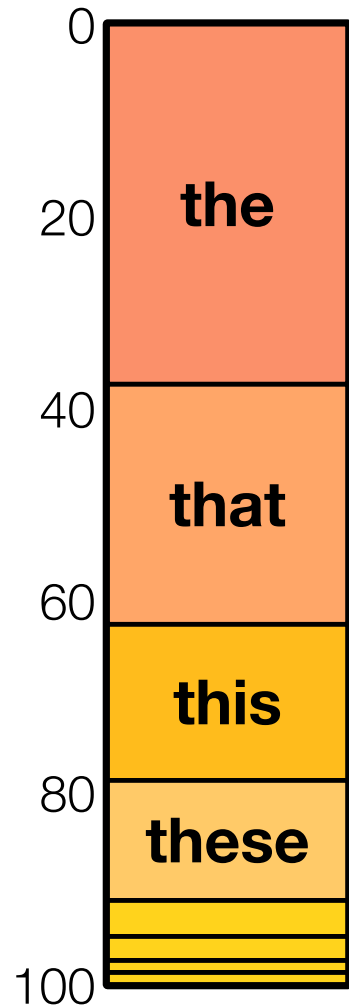
Optimization versus Variety

- Decoding methods geared towards most probably predictions
 - greedy decoding
 - beam search
 - minimum Bayes risk (more on that later)
- A different goal: a variety of responses

→ Random sampling

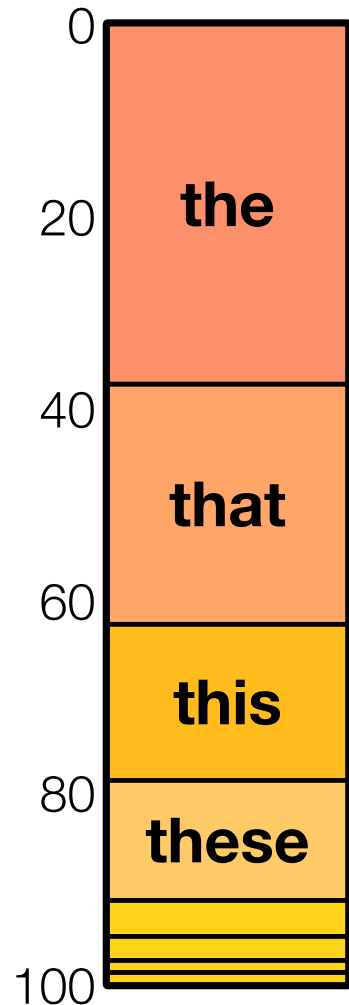
This is commonly used in large language models
(where decoding is called inference)

Core Idea



Pick words randomly
based on
their predicted probability

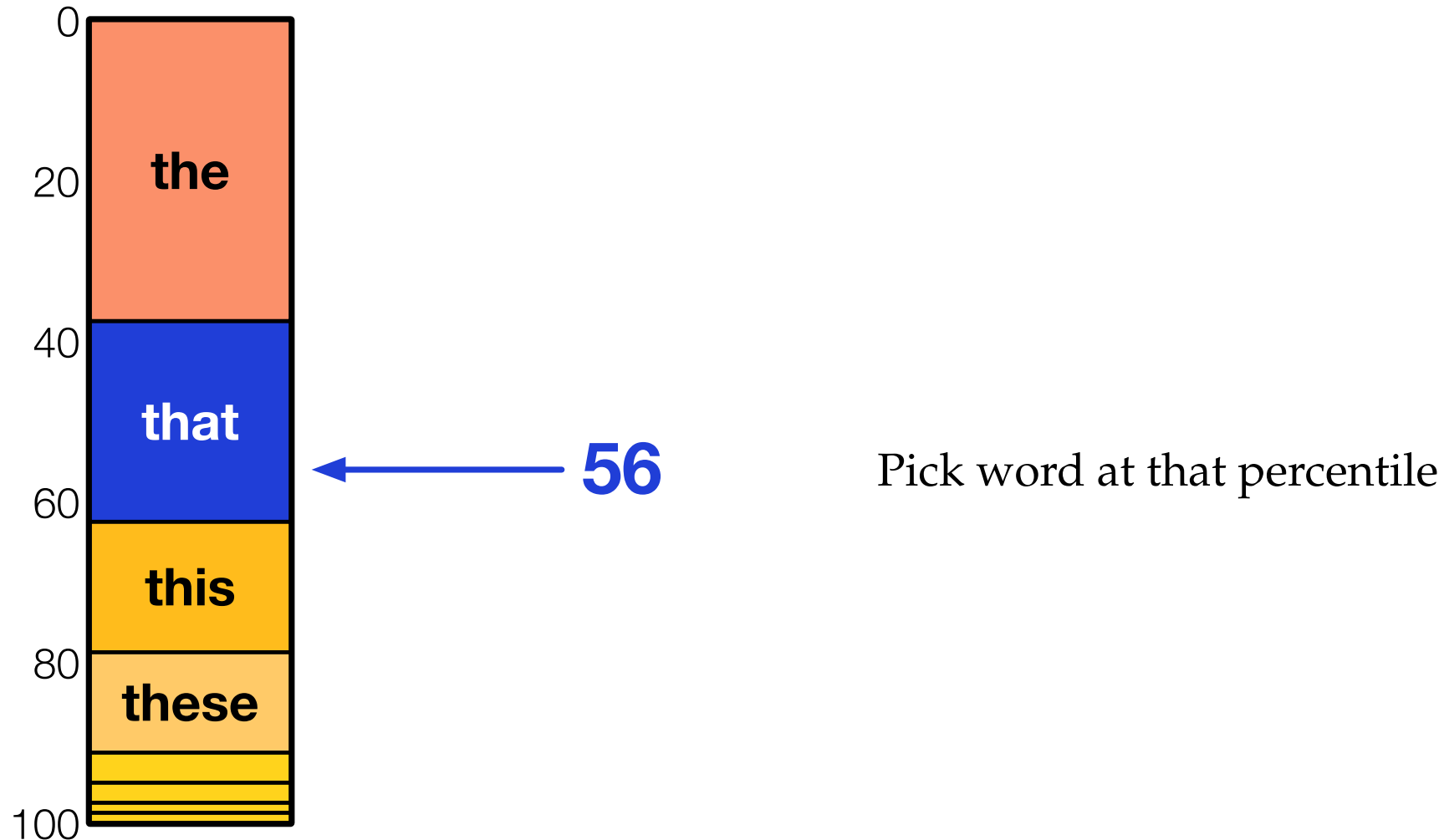
Random Number: 56



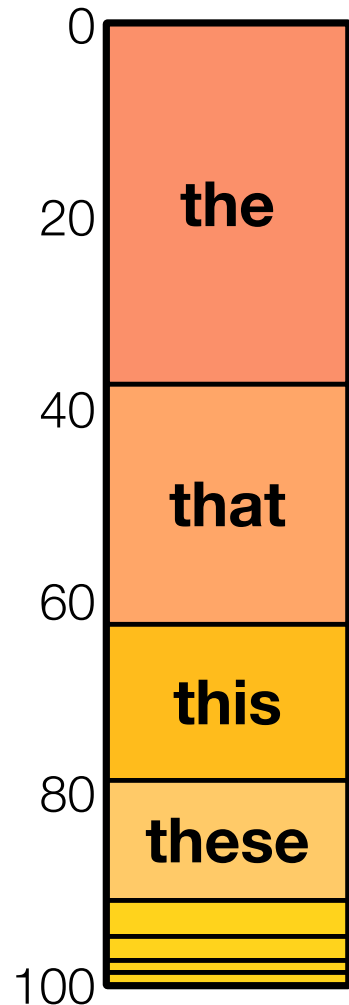
56

Randomly generate a number

Random Number: 56



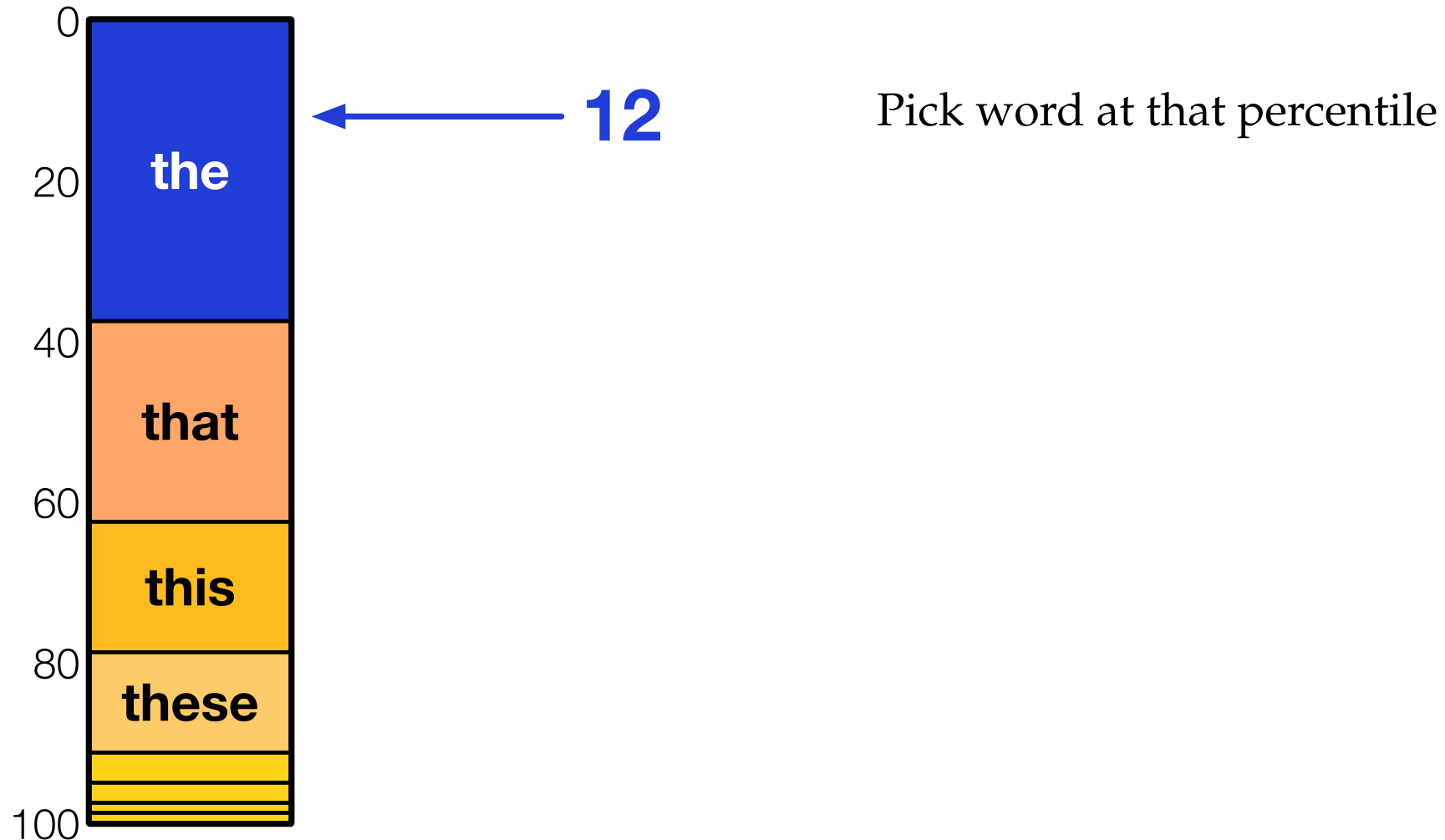
Random Number: 12



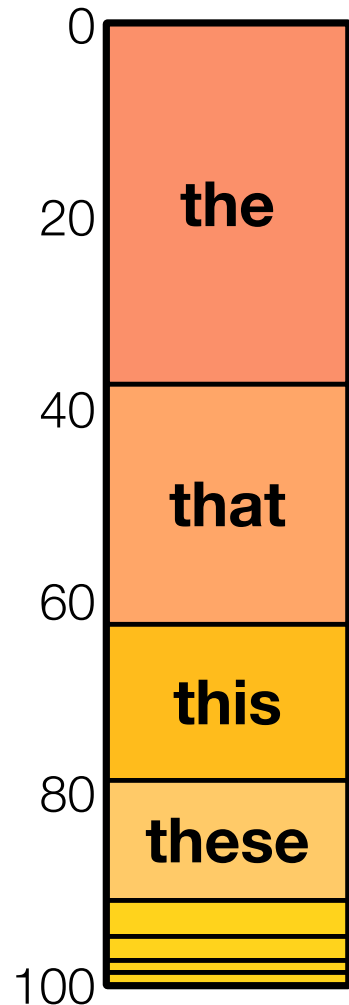
12

Randomly generate a number

Random Number: 12



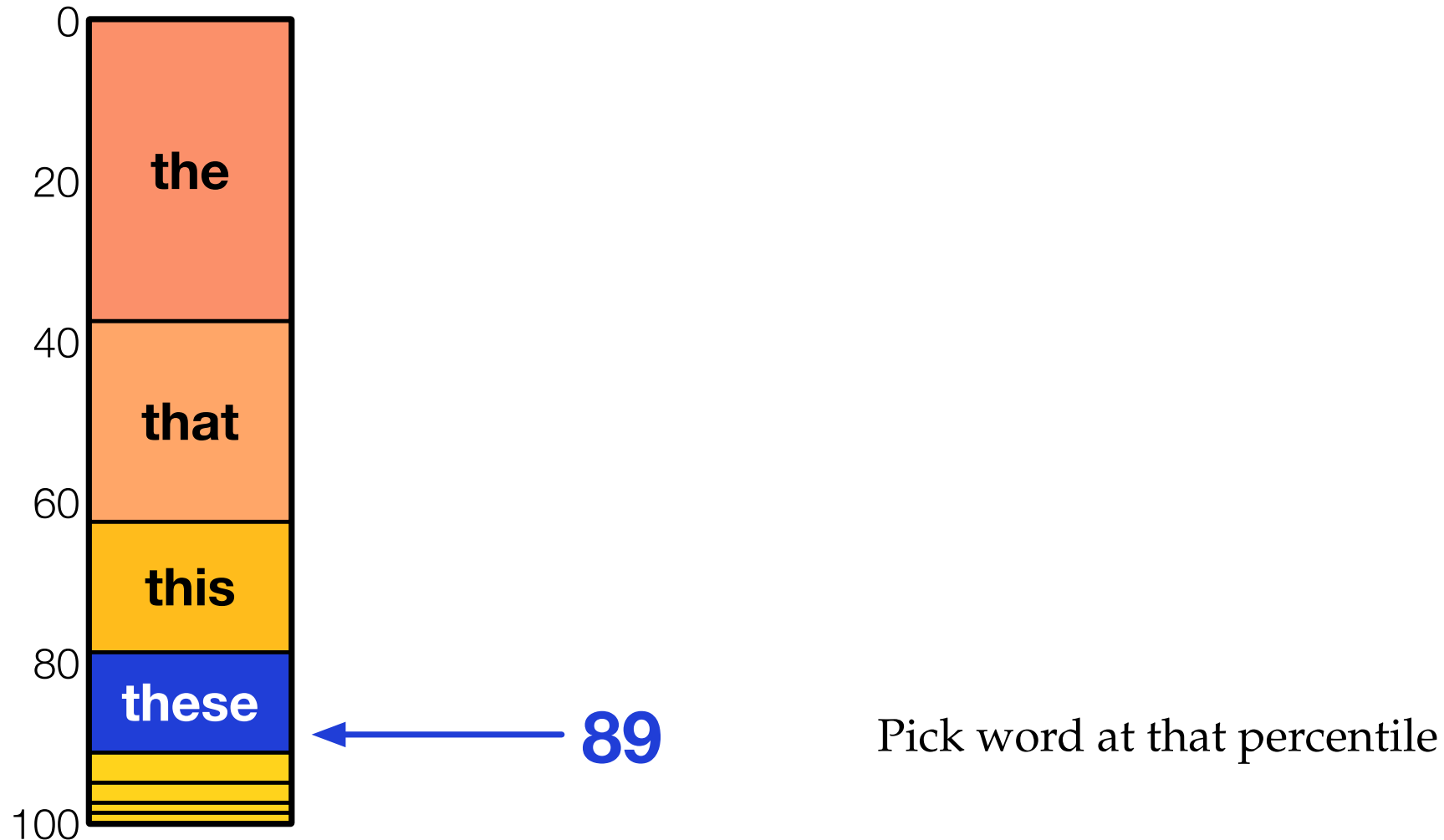
Random Number: 89



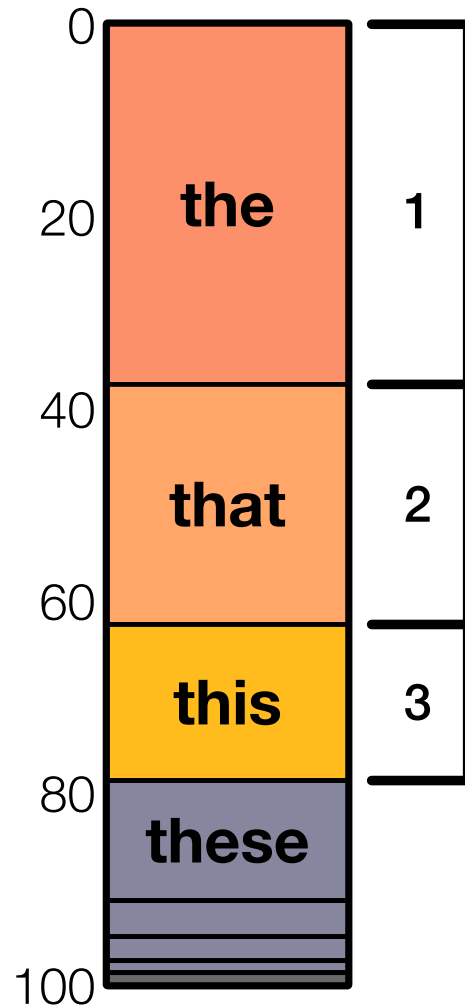
89

Randomly generate a number

Random Number: 89

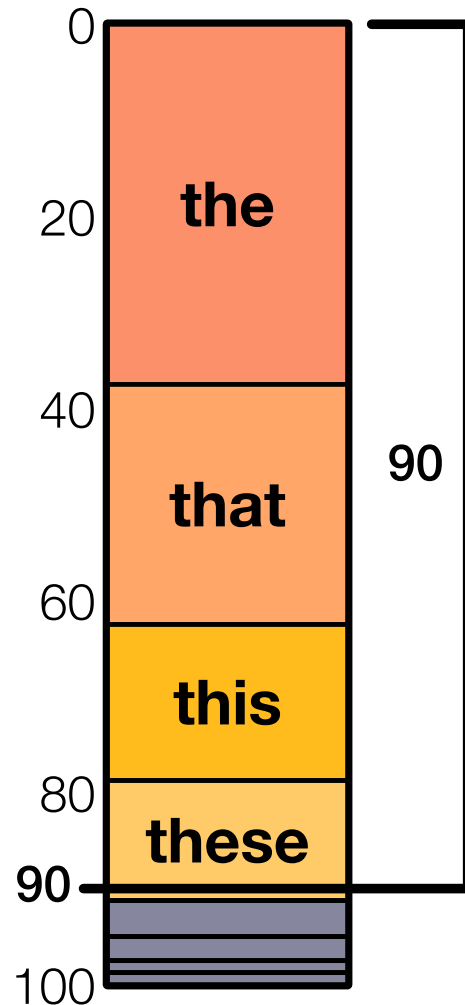


Top-k



Limit to the top $k = 3$ choices

Top-p



Limit to words that fall into
the top $p = 90$ percent

Temperature

- Where do the probabilities come from?
- Recall
 - the prediction layer first computes a number y_i for each word i
 - this number gets normalized with the softmax activation function

$$\text{softmax}(y_i) = \frac{e^{y_i}}{\sum_j e^{y_j}}$$

- Temperature sampling adds a temperature T

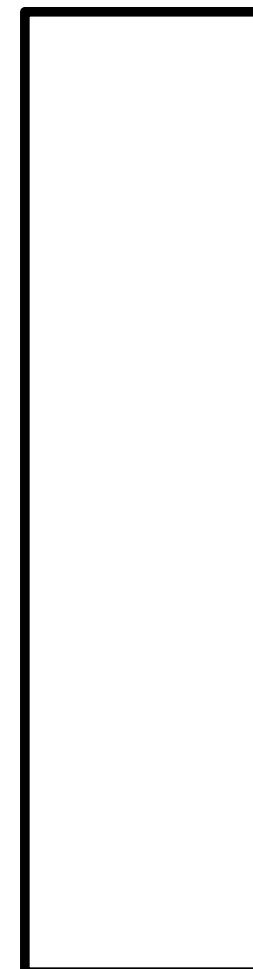
$$\text{softmax}(y_i) = \frac{e^{\frac{y_i}{T}}}{\sum_j e^{\frac{y_j}{T}}}$$

Temperature $T=1$

29



	y_i	$\exp(y_i)$	<i>softmax</i>
the	0.53		
that	0.26		
this	-0.22		
these	-0.51		

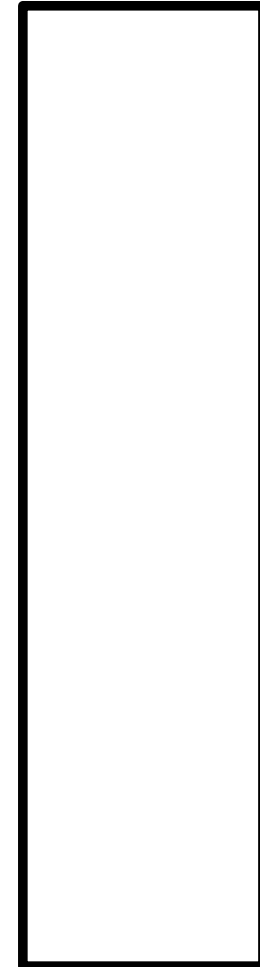


Temperature T=1

30



	y_i	$\exp(y_i)$	<i>softmax</i>
the	0.53	1.7	
that	0.26	1.3	
this	-0.22	0.8	
these	-0.51	0.6	



Temperature $T=1$

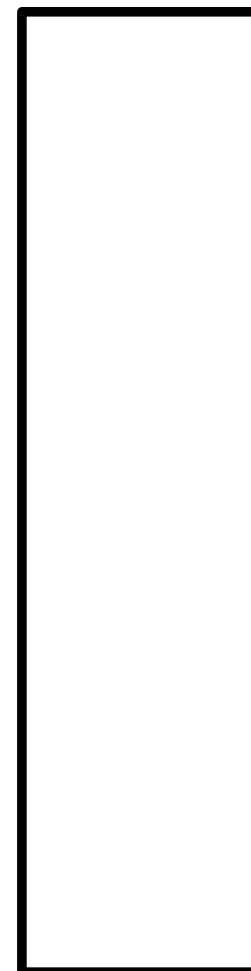


Temperature $T=10$

32



	y_i	$y_i/10$	$\exp(y_i/10)$	<i>softmax</i>
the	0.53			
that	0.26			
this	-0.22			
these	-0.51			

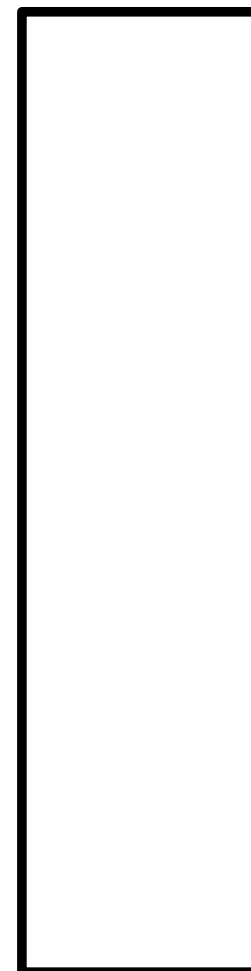


Temperature $T=10$

33



	y_i	$y_i/10$	$\exp(y_i/10)$	<i>softmax</i>
the	0.53	0.053		
that	0.26	0.026		
this	-0.22	-0.022		
these	-0.51	-0.051		

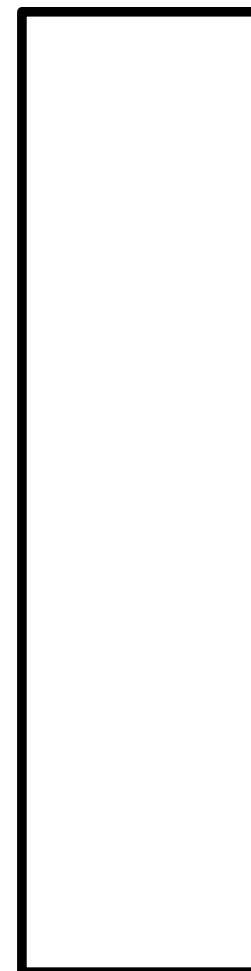


Temperature $T=10$

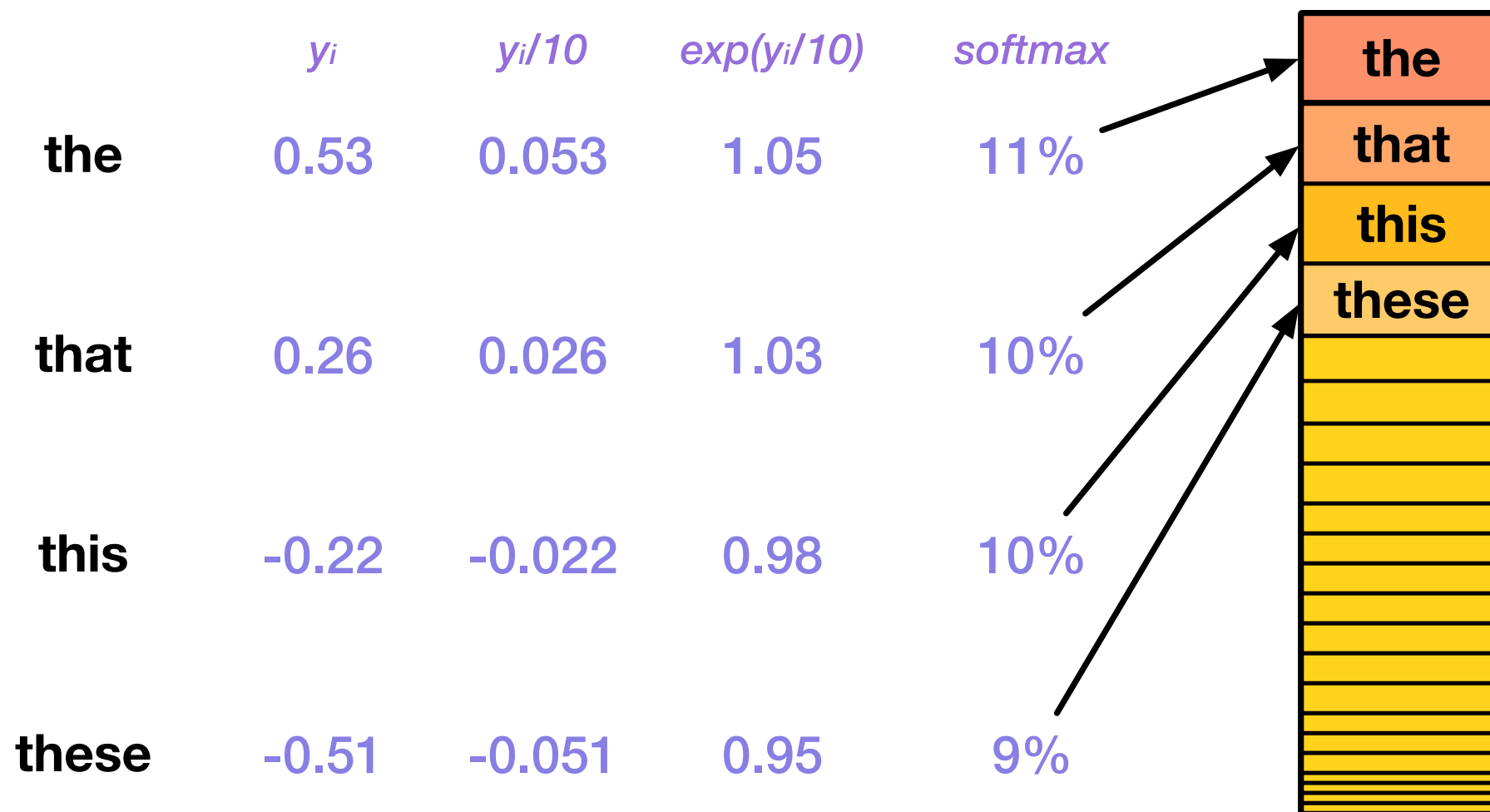
34



	y_i	$y_i/10$	$\exp(y_i/10)$	<i>softmax</i>
the	0.53	0.053	1.05	
that	0.26	0.026	1.03	
this	-0.22	-0.022	0.98	
these	-0.51	-0.051	0.95	



Temperature T=10

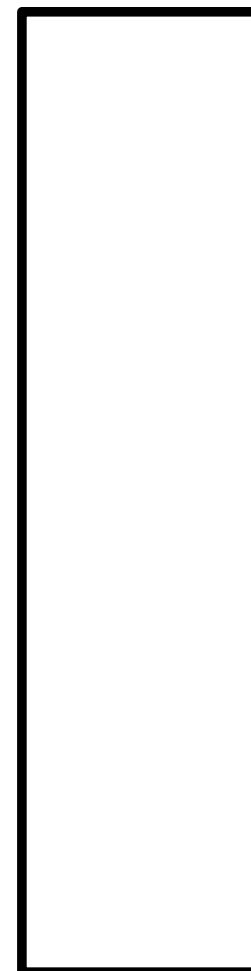


Temperature $T=0.1$

36

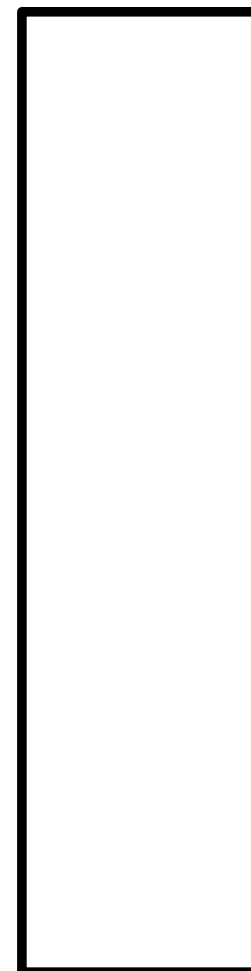


	y_i	$y_i/0.1$	$\exp(y_i/0.1)$	<i>softmax</i>
the	0.53			
that	0.26			
this	-0.22			
these	-0.51			



Temperature $T=0.1$

	y_i	$y_i/0.1$	$\exp(y_i/0.1)$	<i>softmax</i>
the	0.53	5.3		
that	0.26	2.6		
this	-0.22	-2.2		
these	-0.51	-5.1		

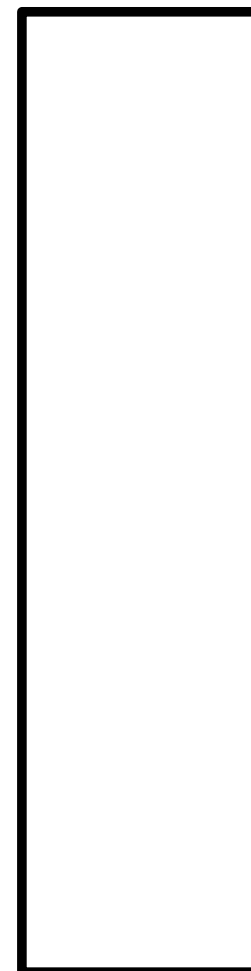


Temperature $T=0.1$

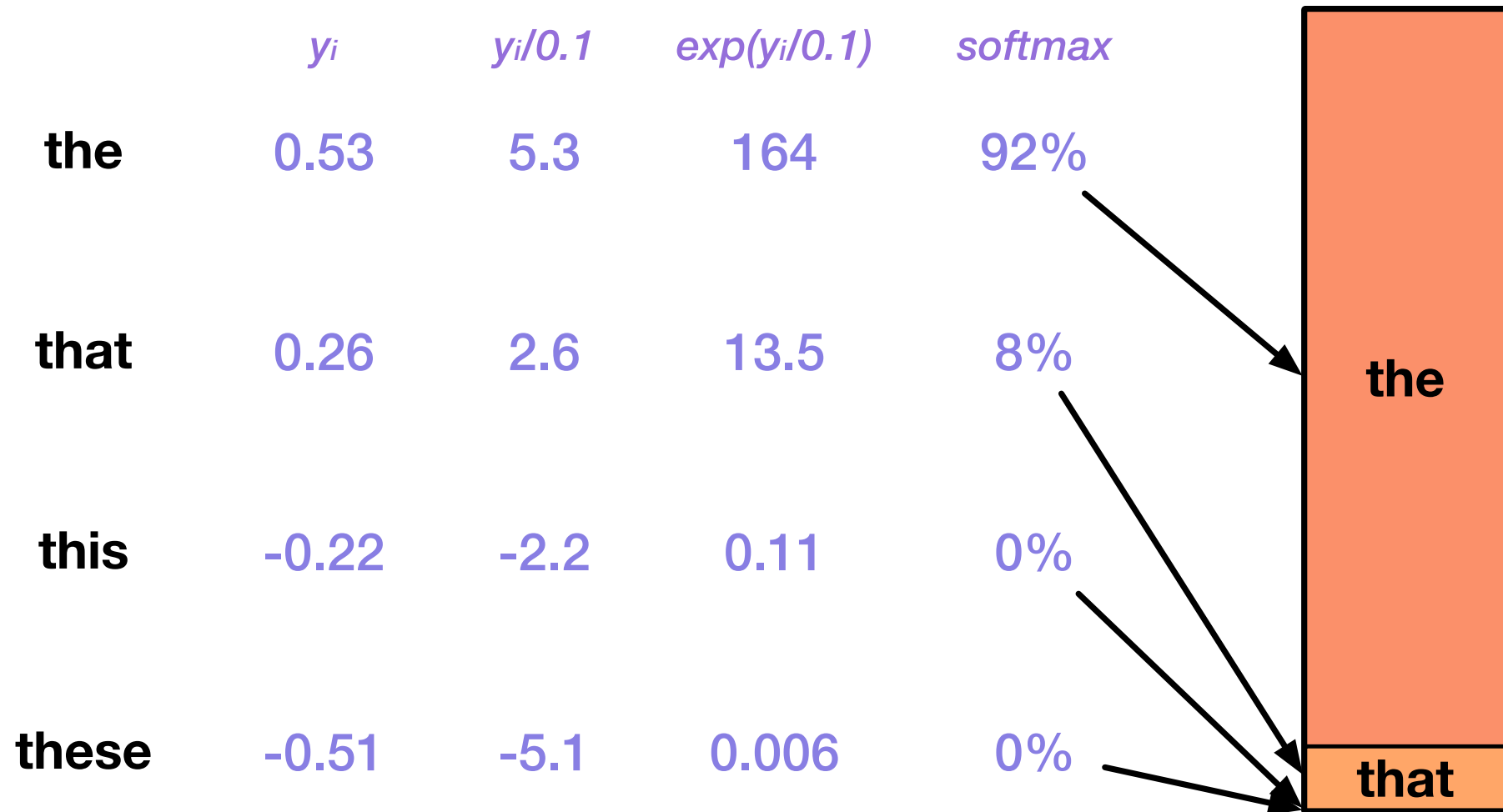
38



	y_i	$y_i/0.1$	$\exp(y_i/0.1)$	<i>softmax</i>
the	0.53	5.3	164	
that	0.26	2.6	13.5	
this	-0.22	-2.2	0.11	
these	-0.51	-5.1	0.006	



Temperature $T=0.1$



ensembling

Ensembling

- Train multiple models
- Say, by different random initializations

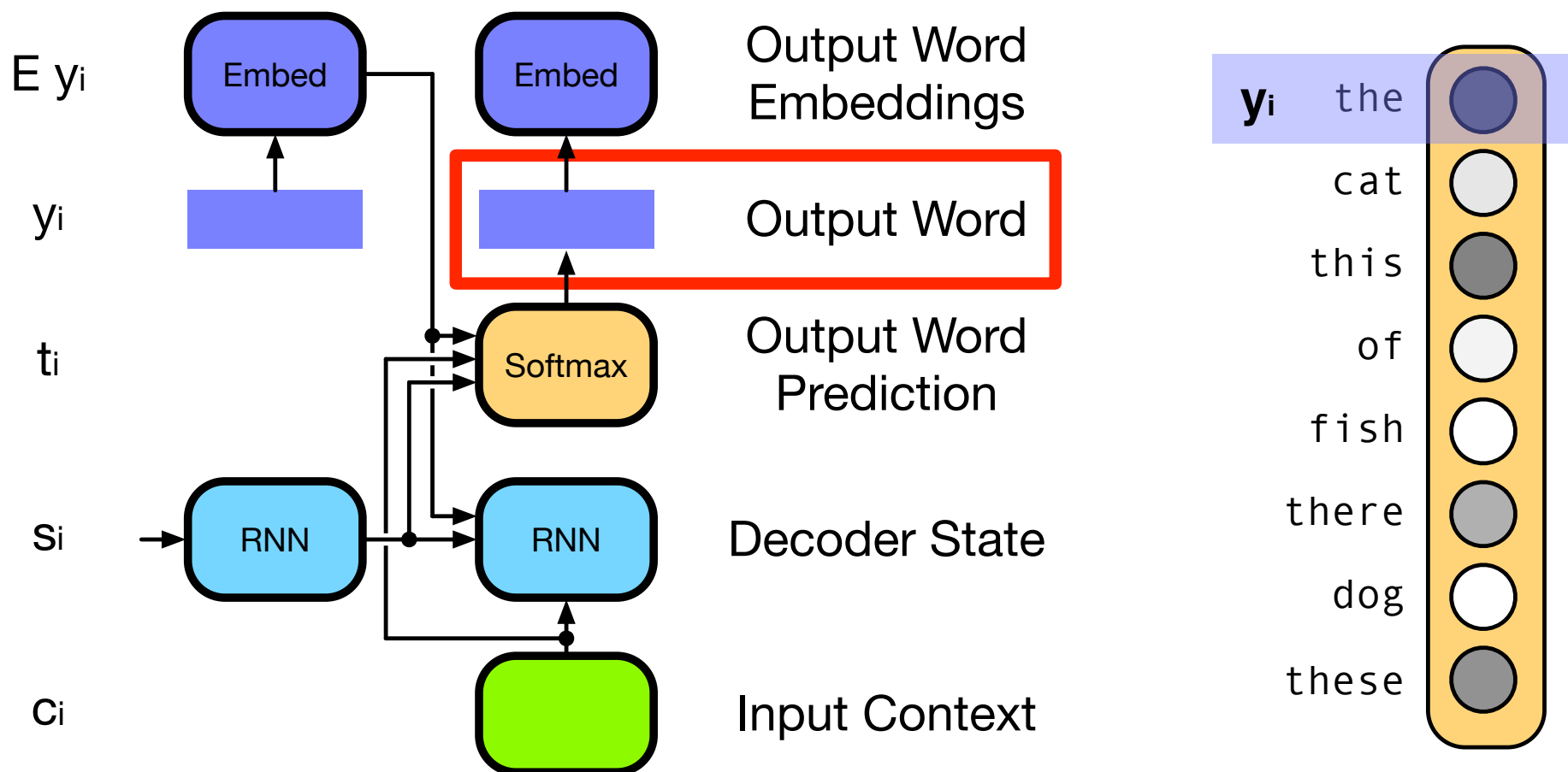


- Or, by using model dumps from earlier iterations

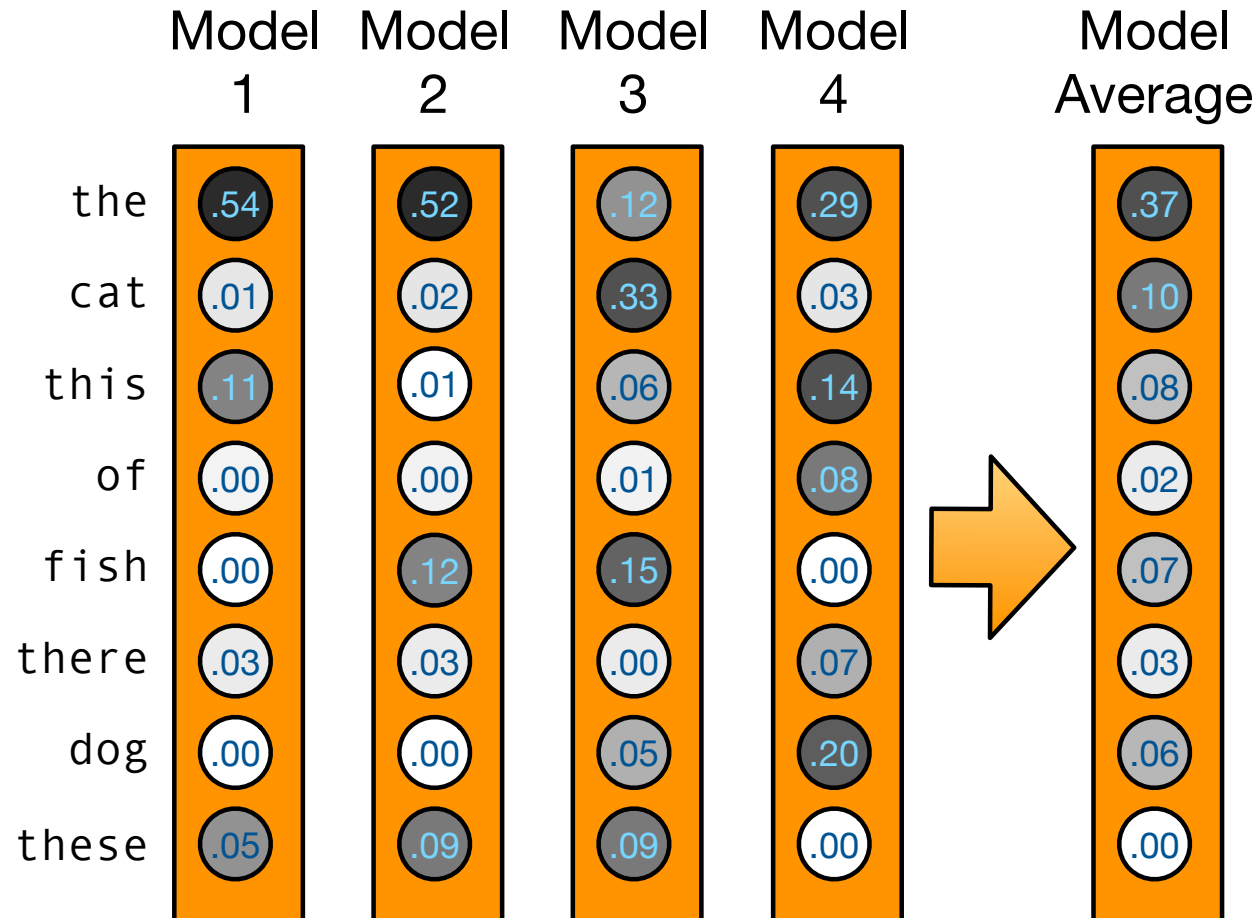


(most recent, or interim models with highest validation score)

Decoding with Single Model



Combine Predictions



Ensembling

- Surprisingly reliable method in machine learning
- Long history, many variants:
bagging, ensemble, model averaging, system combination, ...
- Works because errors are random, but correct decisions unique

reranking

Right-to-Left Inference

- Neural machine translation generates words right to left (L2R)

the → cat → is → in → the → bag → .

- But it could also generate them right to left (R2L)

the ← cat ← is ← in ← the ← bag ← .

Obligatory notice: Some languages (Arabic, Hebrew, ...) have writing systems that are right-to-left, so the use of "right-to-left" is not precise here.

Right-to-Left Reranking



- Train both L2R and R2L model
- Score sentences with both
 - ⇒ use both left and right context during translation

Right-to-Left Reranking

- Train both L2R and R2L model
- Score sentences with both
 - ⇒ use both left and right context during translation
- Only possible once full sentence produced → re-ranking
 1. generate n-best list with L2R model
 2. score candidates in n-best list with R2L model
 3. chose translation with best average score

- Recall: Bayes rule

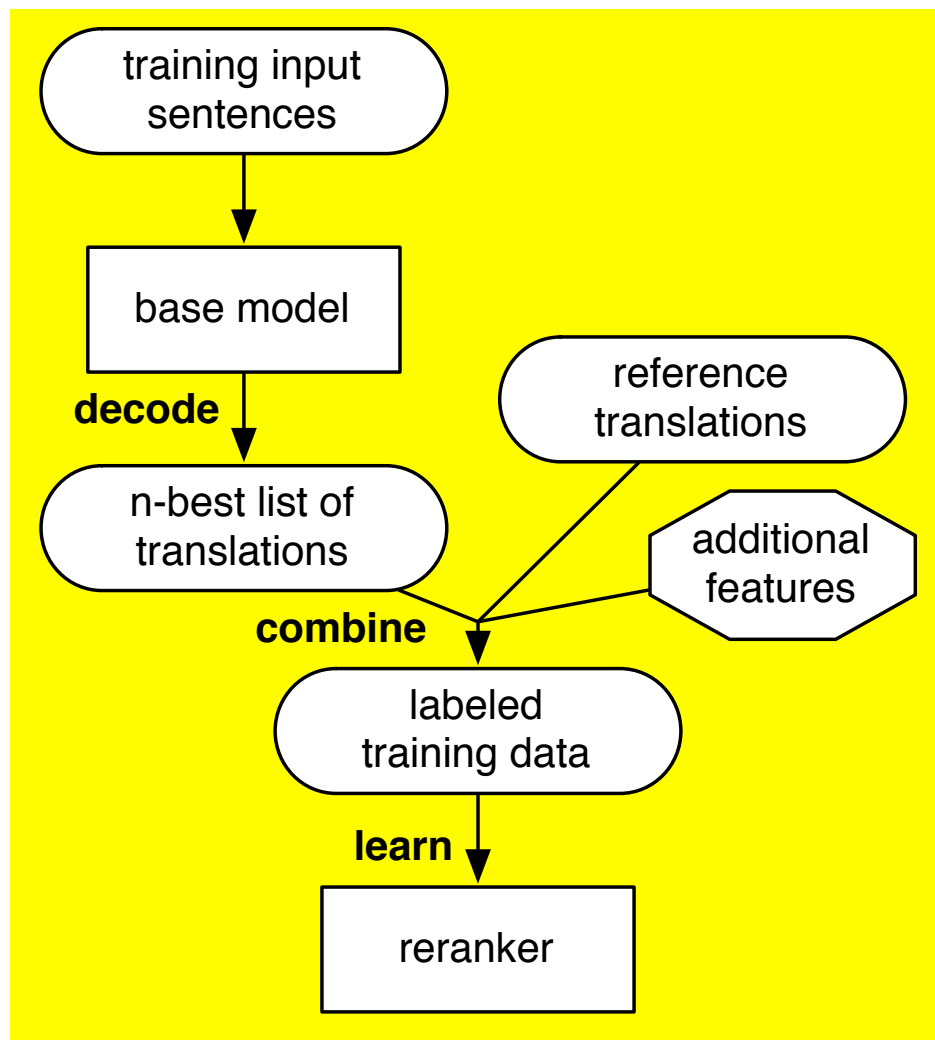
$$p(y|x) = \frac{1}{p(x)} p(x|y) p(y)$$

- Language model $p(y)$
 - trained on monolingual target side data
 - can already be added to ensemble decoding
- Inverse translation model $p(x|y)$
 - train a system in the reverse language direction
 - used in reranking

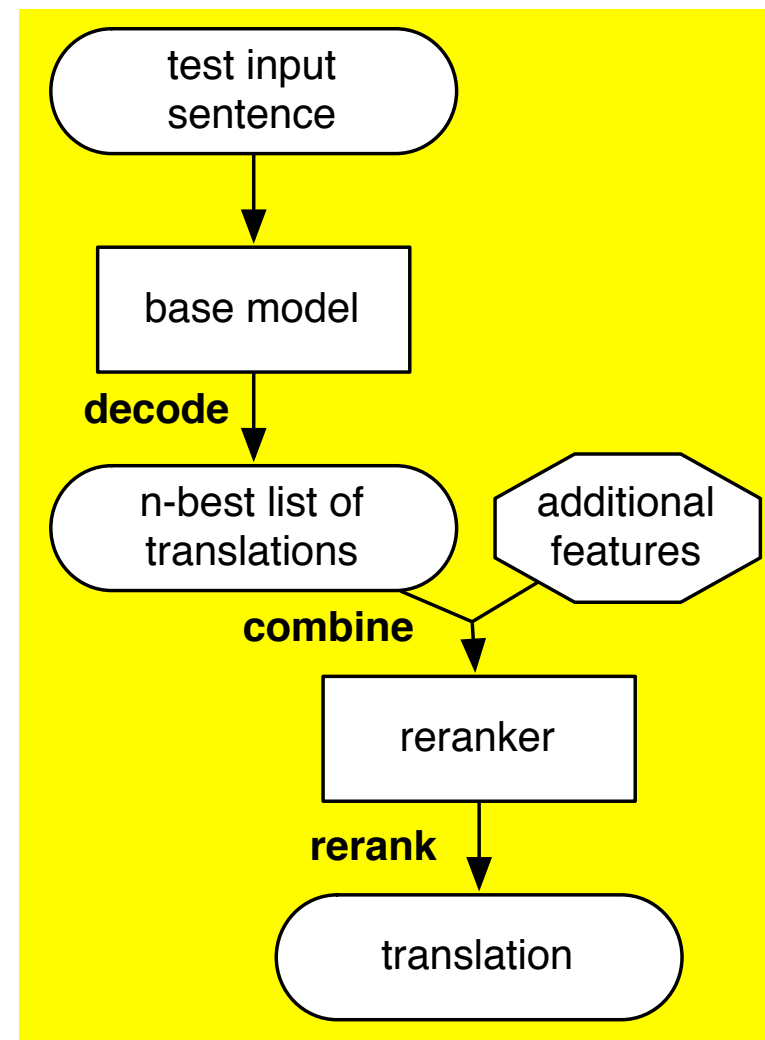
- Several models provide a score each
 - regular model
 - inverse model
 - right-to-left model
 - language model
- These scores could be just added up
- Typically better: weighting the score to optimize translation quality

Training Reranker

Training



Testing



Learning Reranking Weights

- Minimum error rate training (MERT)
 - optimize one weight at a time, leave others constant
 - check how different values change n-best lists
 - only a some threshold values change ranking
 - can be done exhaustively

Learning Reranking Weights

- Minimum error rate training (MERT)
 - optimize one weight at a time, leave others constant
 - check how different values change n-best lists
 - only a some threshold values change ranking
 - can be done exhaustively
- Pairwise Ranked Optimization (PRO)
 - for each sentence in tuning set
 - for each pair of translations in n-best list
 - check which one is a better translation, leaving everything else fixed
 - create a training example
 - (difference in feature values → { better, worse })
 - train linear classifier that learns weights for each feature
- This has not been explored much in neural machine translation

Lack of Diversity

Translations of the German sentence

Er wollte nie an irgendeiner Art von Auseinandersetzung teilnehmen.

He never wanted to participate in any kind of confrontation.

He never wanted to take part in any kind of confrontation.

He never wanted to participate in any kind of argument.

He never wanted to take part in any kind of argument.

He never wanted to participate in any sort of confrontation.

He never wanted to take part in any sort of confrontation.

He never wanted to participate in any sort of argument.

He never wanted to take part in any sort of argument.

He never wanted to participate in any kind of controversy.

He never wanted to take part in any kind of controversy.

He never intended to participate in any kind of confrontation.

He never intended to take part in any kind of confrontation.

He never wanted to take part in some sort of confrontation.

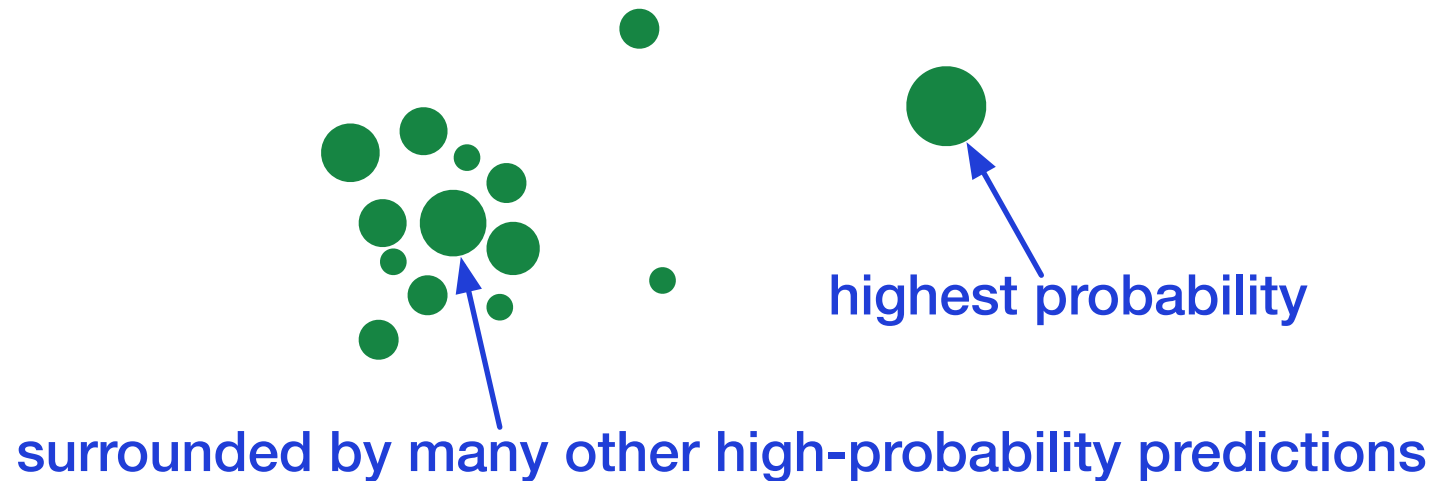
He never wanted to take part in any sort of controversy.

- Monte Carlo decoding
 - Diversity bias term
 - extension of regular beam search
 - add a cost for extending a hypothesis based on rank of word choice
 - * most probable word: no cost
 - * second most probable word: cost c
 - * third most probable word: cost $2c$
- ⇒ prefer to extend many different hypotheses

minimum bayes risk

Motivation

- Predicting a point in space



- Minimum Bayes Risk:
less risky to predict the point in high-probability neighborhood

- Requires a distance function between predictions
- In machine translation: evaluation metrics
 - in SMT days: BLEU score
 - today: COMET scores
- Also required: sampling of many translation for each sentence (very expensive)

Formulation

- We have the probability of each translation: $p(y|x)$
- We have a distance function between any two translations y_1 and y_2
e.g., the COMET score: $\text{COMET}(y_1, y_2)$

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- For each translation, we check how similar it is to other high-scoring translations

$$\sum_{y' \in \mathcal{Y}} p(y'|x) \text{COMET}(y, y')$$

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e.g., the COMET score: $\text{COMET}(y_1, y_2)$
- For each translation, we check how similar it is to other high-scoring translations

$$\sum_{y' \in \mathcal{Y}} p(y'|x) \text{COMET}(y, y')$$

- We pick the one that is scores highest under this measure

$$\hat{y} = \arg \max_{y \in \mathcal{Y}} \sum_{y' \in \mathcal{Y}} p(y'|x) \text{COMET}(y, y')$$

Does it Work?

- Yes — this produces the best translations
- It is very expensive — many samples need to be generated (e.g., 1000)
- It is biased towards translations that are scored highly by the COMET score

[High Quality Rather than High Model Probability: Minimum Bayes Risk Decoding with Neural Metrics, Markus Freitag, David Grangier, Qijun Tan, Bowen Liang, TACL 2022]

constraint decoding

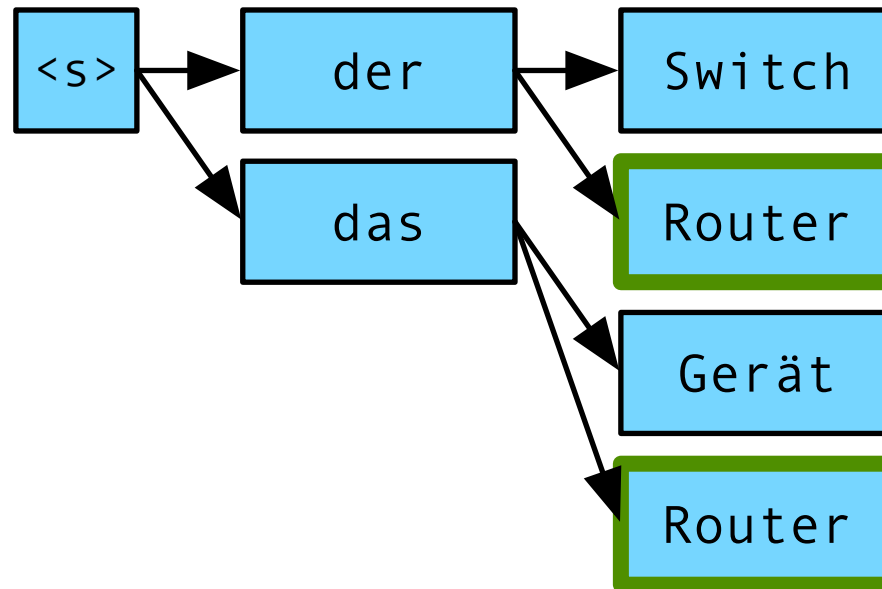
Specifying Decoding Constraints

- Overriding the decisions of the decoder
- Why?
 - ⇒ translations have followed strict terminology
 - ⇒ rule-based translation of dates, quantities, etc.
 - ⇒ interactive translation prediction

The <x translation="Router"> *router* </x> *is* <wall/>
a model <zone> *Psy X500 Pro* </zone> .

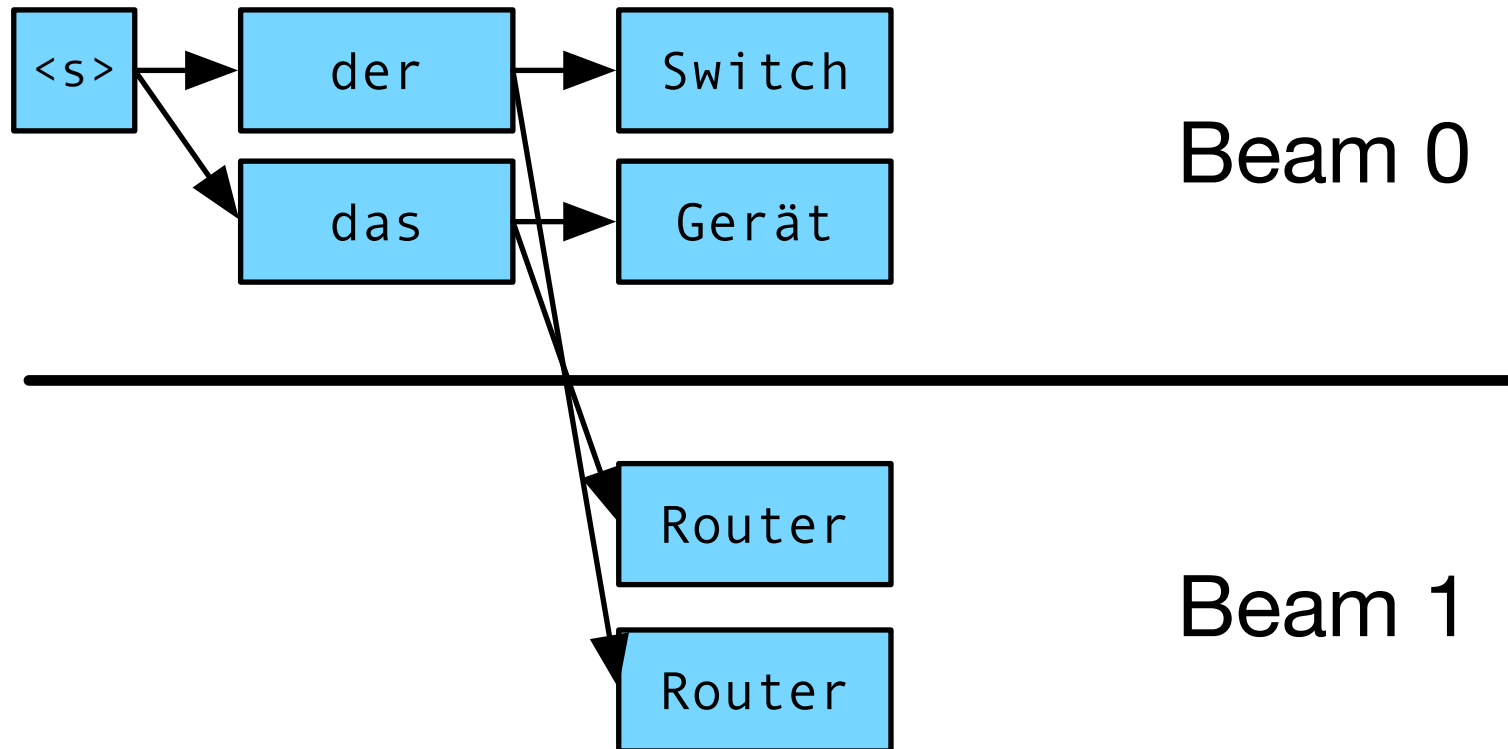
- The XML tags specify to the decoder that
 - the word *router* to be translated as *Router*
 - *The router is*, to be translated before the rest (<wall/>)
 - brand name *Psy X500 Pro* to be translated as a unit (<zone>, </zone>)

The `<x translation="Router"> router </x>` is a model Psy X500 Pro .

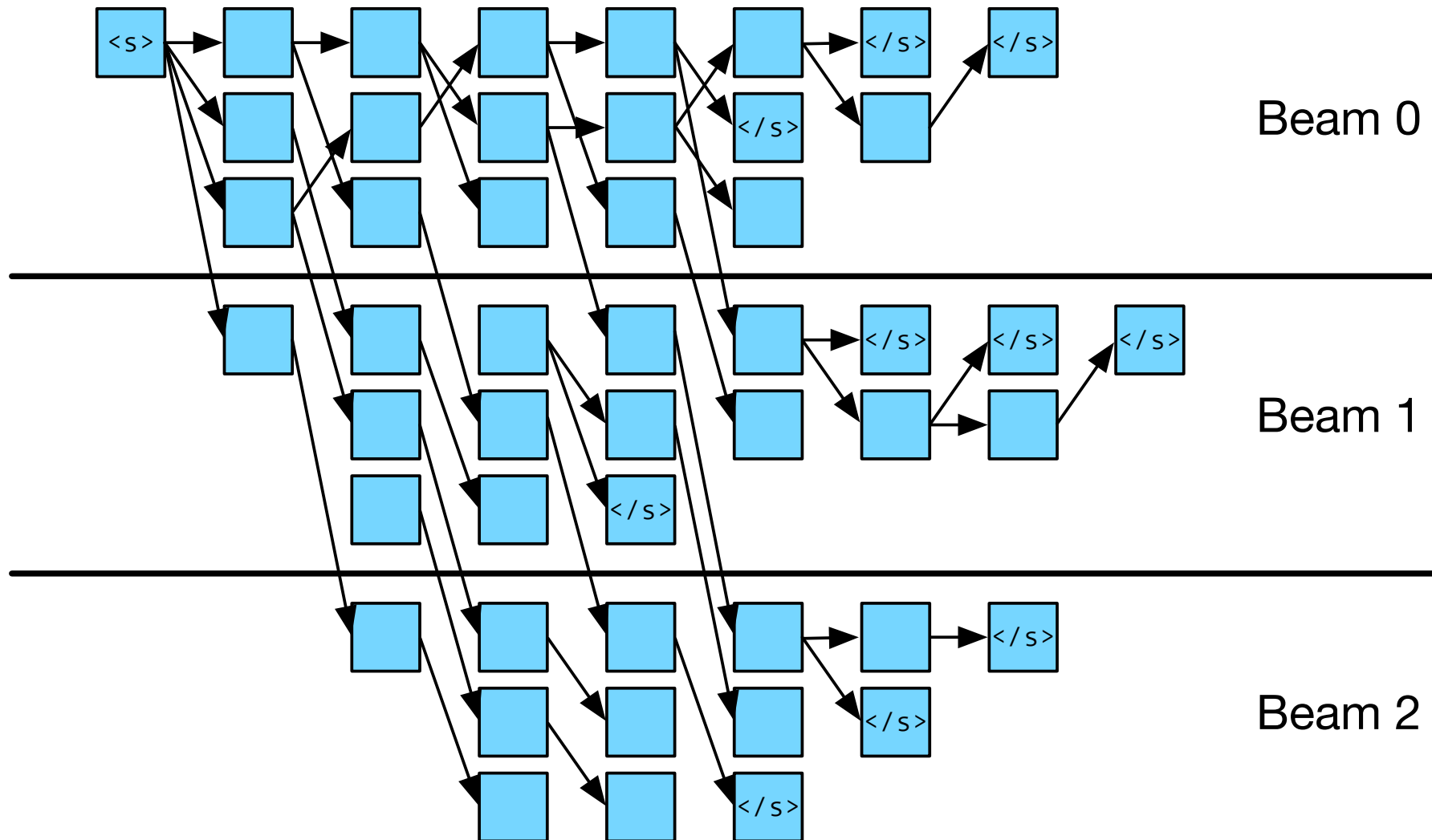


- Satisfying constraints typically costly (overriding model-best choices)
- Solution: separate beam, based on how many constraints satisfied

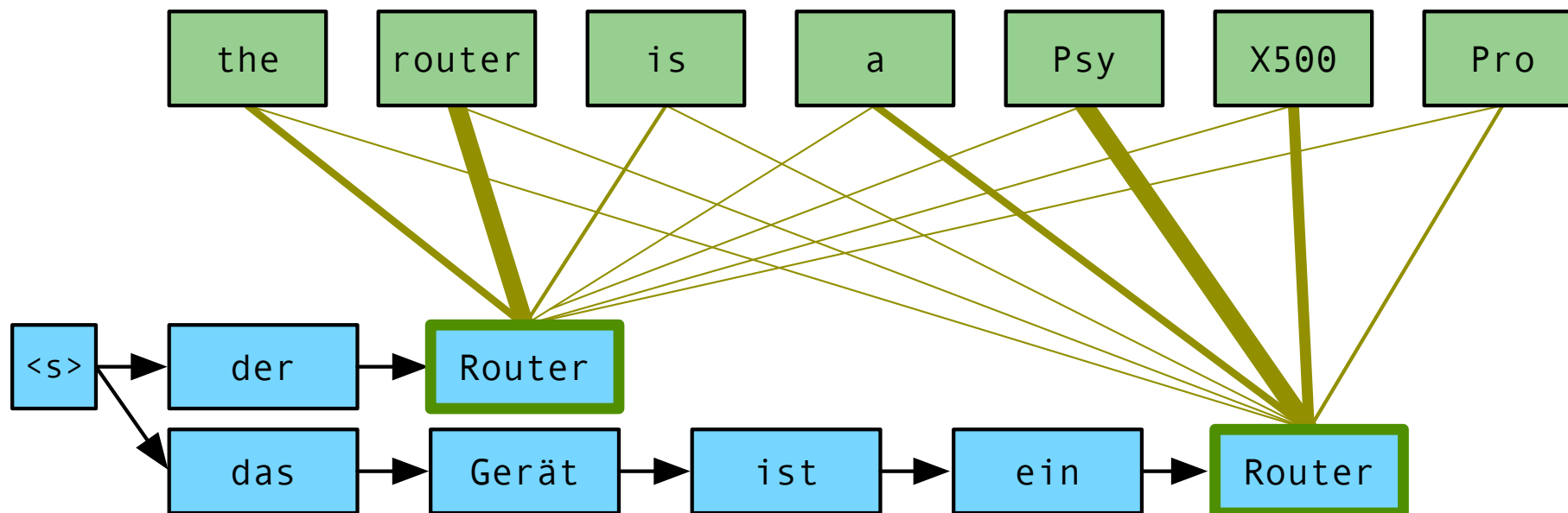
Grid Search



Grid Search



Considering Alignment



- Two hypothesis that fulfill the constraint
 - first one has relevant input words in attention focus
 - second one does not have relevant input words in attention focus

Considering Alignment

- When satisfying a constraint...
 - minimum amount of attention needs to be paid to source
 - use alignment scores as additional cost
- When not satisfying a constraint...
 - block out attention to words not covered by constraint

questions?